

Online Appendix for
“Geospatial Heterogeneity in Inflation:
A Market Concentration Story”
(NOT FOR PUBLICATION)

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A Robustness: NielsenIQ Consumer Panel

As a robustness check, we use the NielsenIQ Consumer Panel to examine the extent to which households make purchases outside their residential MSA. We construct a household-year-level sample from 2006 to 2020 and identify households that make purchases within and outside their residential MSA in each year. Table A.1 shows that 92% of households made purchases exclusively within their residential MSA.

Furthermore, Table A.2 shows that household characteristics and shopping patterns, including income, the number of stores visited, and total spending, are similar across groups. For households that shop outside of their MSA, they visit an average of 1.75 stores, spend approximately 50% of their total expenditure outside their residential MSA, and the average number of these outside MSAs they purchase from is 1.05.

In addition, we compute income deciles using two different MSA definitions in NielsenIQ. One

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Table A.1: Fraction of Households Shopping within their Residential MSA

Indicator	Observation	Percent
1	780,500	92.32
0	64,932	7.68
Total	845,432	100

Notes: The table shows the fraction of households that consume within their residential MSA (with an indicator value of 1) in each given year. The data cover household-year observations from 2006 to 2020.

Table A.2: Characteristics of Households by Shopping Types

Variable	Mean (SD)	Mean (SD)
Indicator	1	0
Income	20.46 (5.98)	19.94 (5.87)
Store #	3.32 (1.90)	3.77 (2.08)
Spending Amount	1812.05 (1985.68)	1659.12 (1811.08)
Store # (out)		1.75 (1.26)
Spending Amount (out)		714.78 (1226.29)
MSA # (out)		1.05 (0.23)
Obs	780,500	64,932

Notes: The table provides the shopping characteristics of households by their types based on whether they shop within their residential MSA (indicator=1) or not (indicator=0). The first column indicates the households only shopping inside their MSA, and the second column shows those shopping outside of their MSA. Store # is the number of stores the households purchase from, Spending Amount is the total amount of spending, Store # (out) is the number of stores outside of the household's living MSA, Spending Amount (out) is the amount of spending made outside of their living MSA, and MSA # (out) is the number of MSAs the household shops, outside of their residential MSA. This is the household-year level sample over 2006-2020.

is based on the MSA information of households in the NielsenIQ Consumer Panel, and the other is based on the MSA information of consumers, derived by linking the locations of stores from which households make purchases in the Scanner data with household income data in the NielsenIQ

Table A.3: Gaps in Two Income Decile Definitions: Household vs. Consumer MSAs

Gap	Observation	Percent
-3	1	0.54
-1	20	10.81
0	140	75.68
1	24	12.97
Total	185	100

Notes: The table computes the gap in income deciles when defined by consumer income and household income, using the MSA-level sample.

Consumer Panel. Table A.3 shows the gap between these two definitions, revealing that most MSAs (75.68%) align with the same income decile definitions, and only a very small fraction (0.54%) exhibit a gap of three deciles. This finding is consistent with our baseline measures of income deciles based on store locations and BEA income per capita data not being mismeasured.

B MSA-level Price Indexes: BLS vs. NielsenIQ

To assess the representativeness of NielsenIQ, we compare NielsenIQ food price indexes with the official CPI series provided by the U.S. Bureau of Labor Statistics for the MSAs available in the CPI data. In particular, we use three MSAs—Chicago, Los Angeles, and New York. NielsenIQ tracks the official series well across all three markets, as shown in Figures B.1, B.2, and B.3.

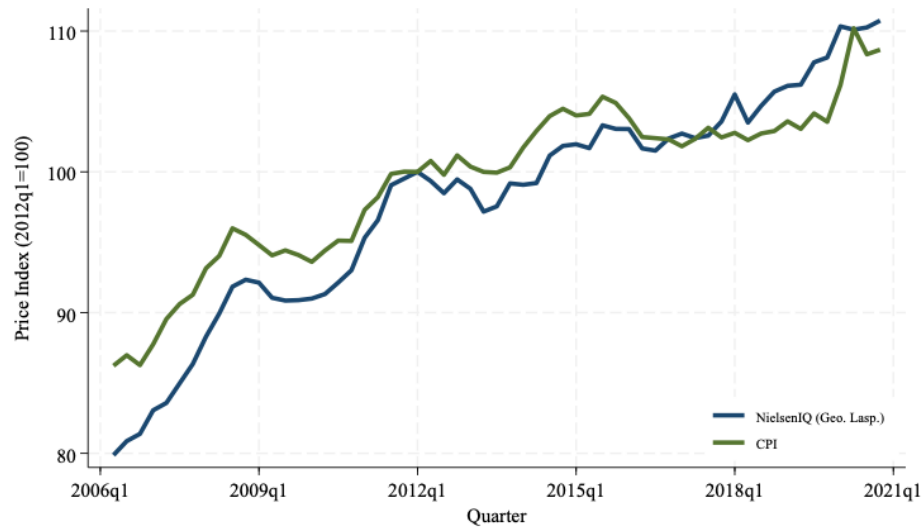


Figure B.1: Price Index for Food and Non-Alcoholic Beverages: CPI vs. NielsenIQ (Chicago MSA)

Notes: This figure compares the BLS CPI with a NielsenIQ Retail Scanner price index for the Chicago MSA. The NielsenIQ index is a Divisia aggregate over 18 PCE items of item-level geometric Laspeyres indices, weighted by average expenditure share in adjacent quarters. Sixteen items enter through their chained indices, while sugar and fruit enter through direct (bilateral) comparisons to the 2006Q2 base quarter, since both chained series suffer from substantial chain drift. No values are imputed. Neither series is seasonally adjusted, and both are normalized to 100 in 2012Q1.



Figure B.2: Price Index for Food and Non-Alcoholic Beverages: CPI vs. NielsenIQ (LA MSA)

Notes: This figure compares the BLS CPI with a NielsenIQ Retail Scanner price index for the Los Angeles MSA. The NielsenIQ index is a Divisia aggregate over 18 PCE items of item-level geometric Laspeyres indices, weighted by average expenditure share in adjacent quarters. Sixteen items enter through their chained indices, while sugar and fruit enter through direct (bilateral) comparisons to the 2006Q2 base quarter, since both chained series suffer from substantial chain drift. No values are imputed. Neither series is seasonally adjusted, and both are normalized to 100 in 2012Q1.

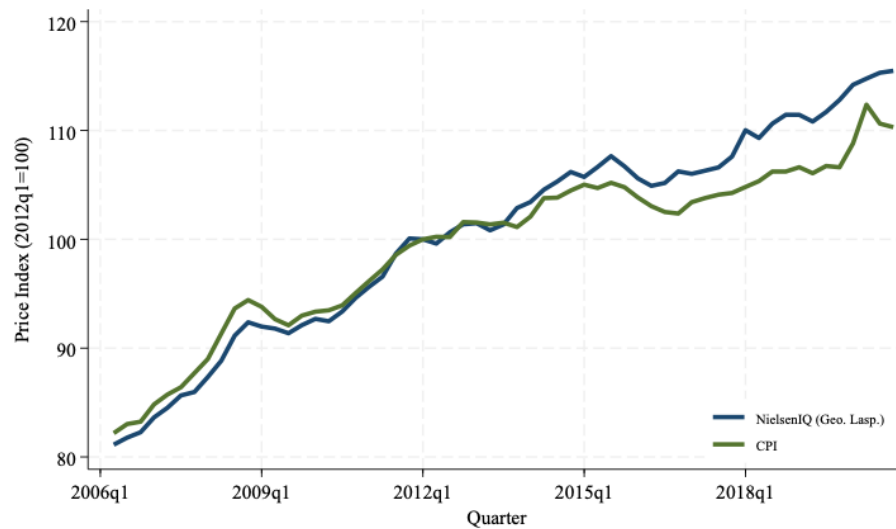


Figure B.3: Price Index for Food and Non-Alcoholic Beverages: CPI vs. NielsenIQ (New York MSA)

Notes: This figure compares the BLS CPI with a NielsenIQ Retail Scanner price index for the New York MSA. The NielsenIQ index is a Divisia aggregate over 18 PCE items of item-level geometric Laspeyres indices, weighted by average expenditure share in adjacent quarters. Sixteen items enter through their chained indices, while sugar and fruit enter through direct (bilateral) comparisons to the 2006Q2 base quarter, since both chained series suffer from substantial chain drift. No values are imputed. Neither series is seasonally adjusted, and both are normalized to 100 in 2012Q1.

C Robustness: Price and Inflation Patterns

C.1 Alternative Price Indexes

A natural concern with Figure 1 is that our central finding, that the poorest decile experiences higher food inflation, may reflect some properties of the geometric Laspeyres formula rather than differences in underlying price changes. Because a Laspeyres index holds expenditure shares fixed at their base-period values, it does not accommodate substitution toward goods whose relative prices fall and may therefore overstate changes in the cost of living (Diewert, 1976). In addition, when constructed as a chained index, it may be subject to chain drift. We therefore rebuild the aggregate food index using three alternative approaches that relax different assumptions: a Tornqvist index (Figure C.1), a weighted time-product dummy (WTPD) index (Figure C.2), and a Sato-Vartia index (Figure C.3).

Figure C.1 reports two Tornqvist series for the bottom and top income deciles: a chained index and a direct (bilateral) index. The Tornqvist index is a superlative price index that incorporates substitution behavior of consumers by using expenditure-share information from both comparison periods (Diewert, 1976). Under the chained index, the gap between deciles persists at 3 index points by the end of the sample. Chained superlative indexes constructed from scanner data are, however, vulnerable to chain drift: high-frequency promotional cycles combined with asymmetric quantity responses can cause the index to ratchet away from its true level as successive links accumulate (Ivancic et al., 2011; de Haan and van der Grient, 2011). Comparing each quarter directly with the base period avoids this issue by construction, since the resulting index no longer depends on the sequence of intermediate-period links. Doing so raises measured inflation for both deciles and widens the gap slightly, to about 3.5 index points.

One limitation of a bilateral (direct) price index is that each comparison is restricted to goods present in both the base quarter and the comparison quarter, so the matched sample erodes as the two move apart. An alternative method to deal with chain drift is to use a multilateral price index such as the WTPD index. The WTPD avoids chain drift by construction while exploiting all available observations, including goods that enter and exit the sample (Diewert, 2005; de Haan and Krsinich, 2014). Figure C.2 plots the WTPD index for the bottom, median, and top deciles. The gap remains at approximately 3.9 index points.

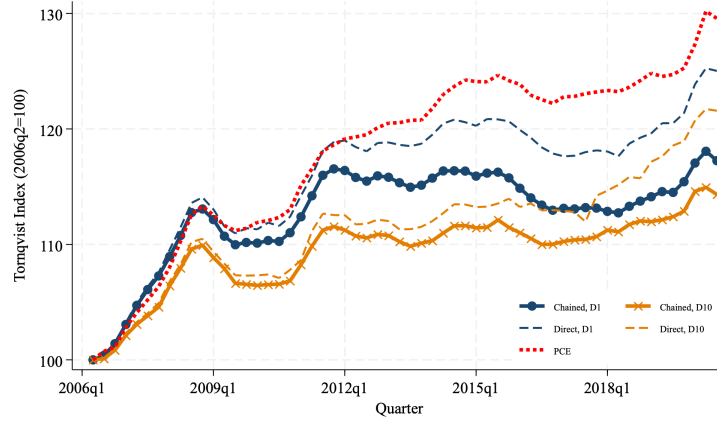


Figure C.1: Tornqvist for Aggregated Food (Chain Drift)

Notes: This figure represents the chained Tornqvist price index and bilateral Tornqvist price index constructed using NielsenIQ Retail Scanner data as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed red line) for aggregate food and beverages. The blue series represents decile 1, the poorest decile of MSAs in our sample, while the gold series represents decile 10, the richest decile of MSAs in our sample. Among these four series, the dashed lines represent direct (bilateral) price indexes and the solid lines represent chained price indexes.

Taken together, the sign and ordering of the inter-decile gap are robust to the choice of price index, though its magnitude varies: the superlative and multilateral indexes yield gaps roughly half as large as that implied by the geometric Laspeyres index. Since the difference largely reflects the substitution margin that these alternative indexes are designed to capture, we treat the multilateral estimate as our conservative benchmark and report the geometric Laspeyres as a fixed-basket comparison.

As a further robustness check, we construct the Sato-Vartia price index, which is exact for a CES cost function (Sato, 1976; Vartia, 1976). Specifically, we replace the weights in equation (1), ω_{mkt} , with

$$\omega_{mkt} = \frac{\frac{(s_{mkt} - s_{mkt-1})}{(\ln s_{mkt} - \ln s_{mkt-1})}}{\sum_{k \in \mathbb{C}_{m,t-1,t}} \frac{(s_{mkt} - s_{mkt-1})}{(\ln s_{mkt} - \ln s_{mkt-1})}},$$

where $\mathbb{C}_{m,t-1,t}$ denotes the set of goods sold in MSA m in both periods. Each price relative is weighted by the logarithmic mean of its expenditure shares in the two periods, normalized to sum to one. Thus, the Sato-Vartia index responds to demand shifts across periods rather than holding the basket fixed at base-period values. Figure C.3 shows that our baseline result remains robust under this substitution-sensitive index.

Lastly, we construct decile-level price indexes by aggregating the MSA-level indexes within

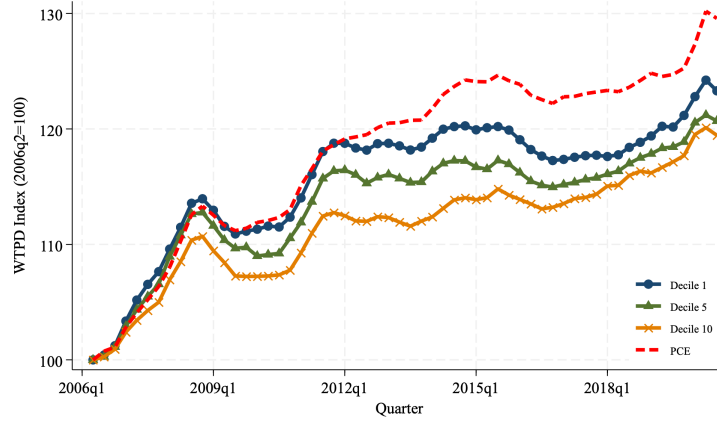


Figure C.2: Weighted Time-Product Dummy (WTPD, Multilateral Index) for Aggregated Food

Notes: This figure represents the weighted time-product dummy (WTPD) index constructed using NielsenIQ Retail Scanner data (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for aggregate food and beverages. Decile 1 corresponds to the poorest decile of MSAs in our sample and decile 10 corresponds to the richest MSAs in our sample.

each income decile. Each MSA is assigned to a decile of income per capita as in Figure C.4, and the aggregation is performed on quarterly log growth rates rather than on index levels: for decile d in quarter t we compute

$$\Delta \ln P_{dt} = \sum_{m \in d} \omega_{mt} \Delta \ln P_{mt}, \quad (\text{C.1})$$

and cumulate the result, normalizing the index to 100 in 2006Q2.¹ The MSA-level growth rates $\Delta \ln P_{mt}$ aggregate the item-level indexes within each MSA.² The weights are Divisia sales shares, $\omega_{mt} = \frac{1}{2} (s_{mt} + s_{m,t-1})$, where s_{mt} is MSA m 's share of decile food sales in quarter t , the same weighting we use to aggregate across items. As shown in Figure C.4, the results based on this approach are consistent with our main findings.³

¹MSA indexes sit at very different levels, so weighting levels by contemporaneous sales shares would let spurious entry and exit in the NielsenIQ store panel move the decile index absent any price change. Weighting growth rates confines panel churn to a reweighting of inflation rates.

²Among the 21 food categories, sugar and fruit exhibit substantial chain drift, which can spuriously affect the trend in aggregate food prices. For these two categories, we therefore use direct bilateral geometric Laspeyres indexes relative to 2006Q2 rather than chained indexes. In addition, we are unable to construct the bilateral geometric Laspeyres index for fruit in 11 of the 185 MSAs. For these MSAs, we impute both the fruit price growth rate and expenditure weight using the median values among the other MSAs in the same income decile.

³The results are also robust when constructing aggregate food indexes without sugar and fruit.

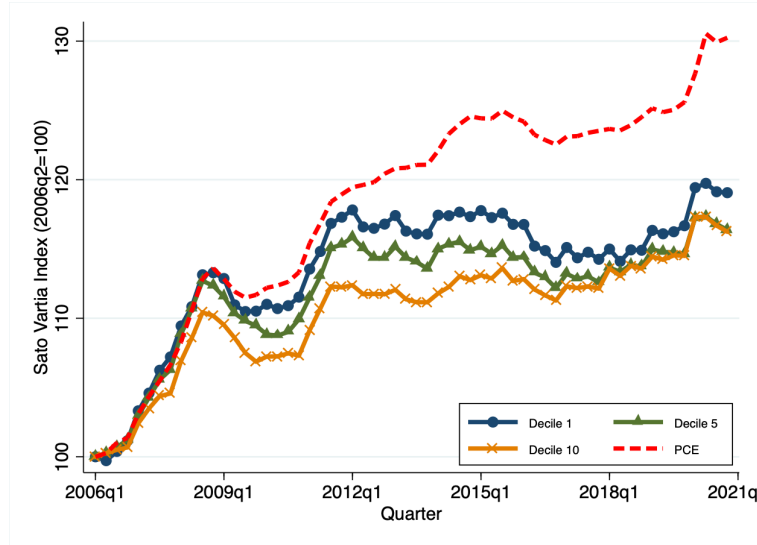


Figure C.3: Demand-based Price Index (Sato-Vartia) for Aggregated Food

Notes: This figure represents the chained Sato-Vartia price index constructed using NielsenIQ Retail Scanner data (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for aggregate food and beverages. All descriptions remain the same as before.

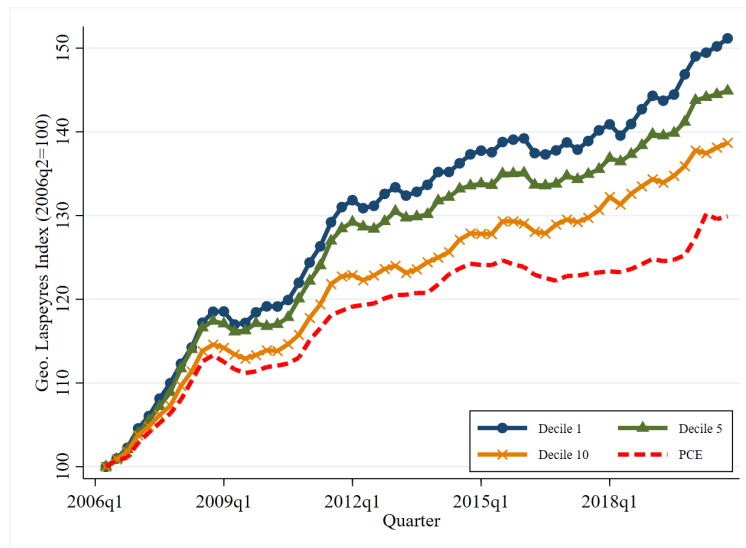


Figure C.4: Price Index for Aggregate Food (MSA-level)

Notes: This figure plots the chained geometric Laspeyres price index constructed from NielsenIQ Retail Scanner data (solid lines) and the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for aggregate food and beverages. All descriptions remain the same as before, except that the MSA-level indexes are aggregated within each income decile by weighting quarterly growth rates using Divisia expenditure shares and cumulating them over time. Fruit and sugar enter through bilateral indexes due to substantial chain drift. A bilateral fruit index cannot be constructed for 11 of the 185 MSAs. For these MSAs, we impute the fruit expenditure weight and growth factor using the medians of the corresponding values among the other MSAs in the same income decile.

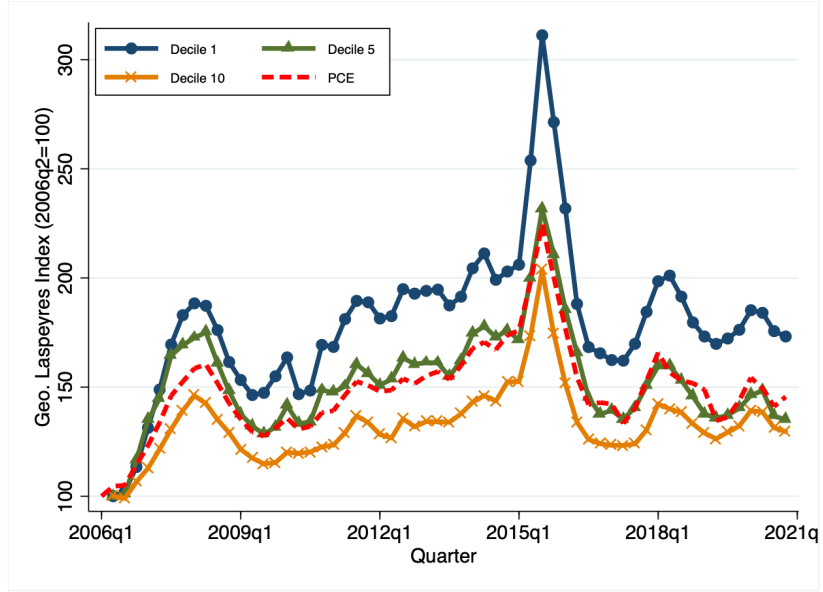


Figure C.5: Price Index for Eggs

Notes: This figure represents the chained geometric Laspeyres price index constructed in the NielsenIQ (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for eggs. All else remains the same as before.

C.2 Other Food Items

We also construct these price indexes for more disaggregated food categories and find consistent results across them. For illustrative purposes, we present results for the following three representative categories: eggs, dairy, and fats and oils.

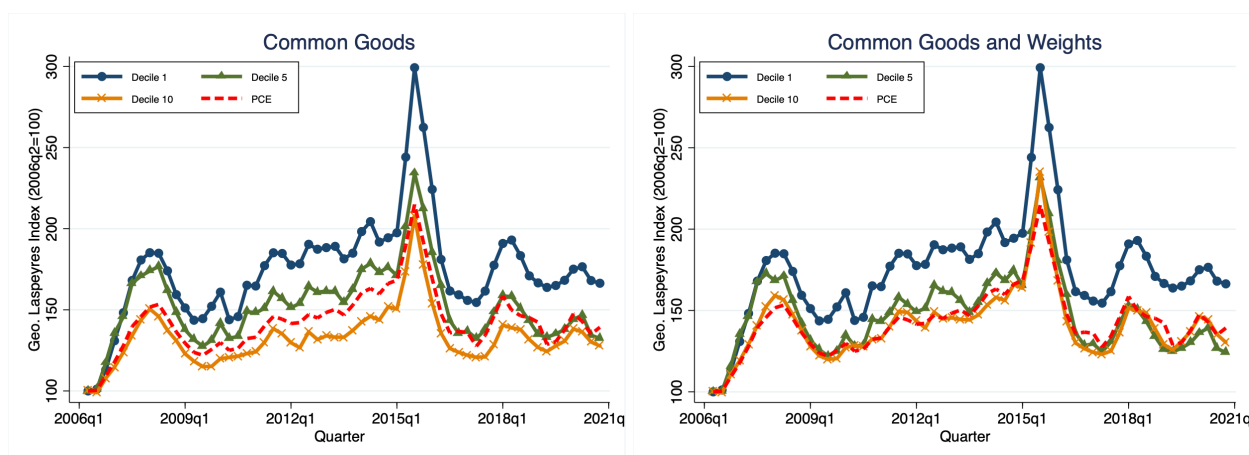


Figure C.6: Price Index for Eggs (Common Goods and Weights)

Notes: This figure represents the chained geometric Laspeyres price index constructed in the NielsenIQ (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for eggs. All descriptions remain the same as before except that the solid lines constructed in NielsenIQ are restricted to the set of goods present across all ten deciles in quarters $t - 1$ and t (named “common goods”) in the left panel, and are restricted to these common goods and further based on the same sales weights in decile 1 in the right panel.

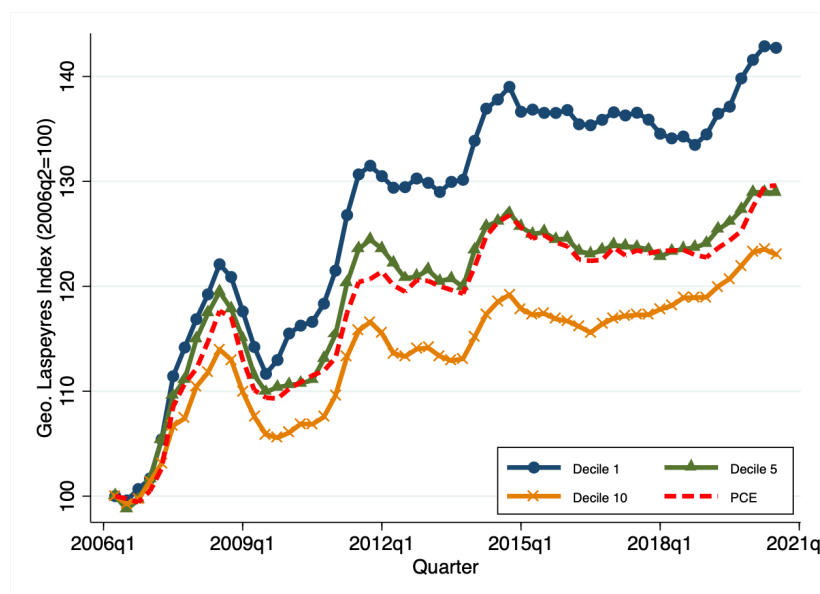


Figure C.7: Price Index for Dairy

Notes: This figure represents the chained geometric Laspeyres price index constructed in the NielsenIQ (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for dairy. All else remains the same as before.

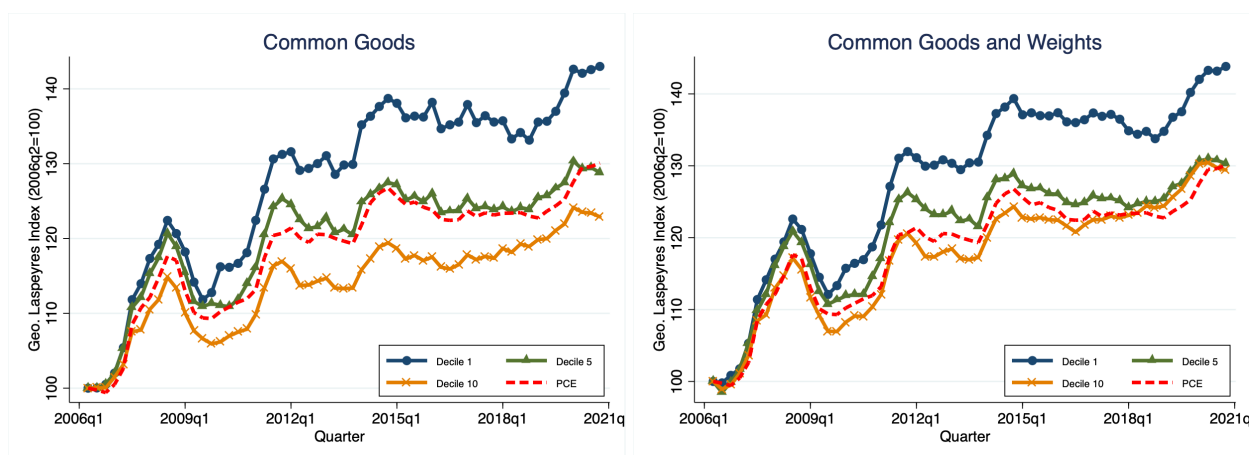


Figure C.8: Price Index for Dairy (Common Goods and Weights)

Notes: This figure represents the chained geometric Laspeyres price index constructed in the NielsenIQ (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for dairy. All descriptions remain the same as before except that the solid lines constructed in NielsenIQ are restricted to the set of goods present across all ten deciles in quarters $t - 1$ and t (named “common goods”) in the left panel, and are restricted to these common goods and further based on the same sales weights in decile 1 in the right panel.

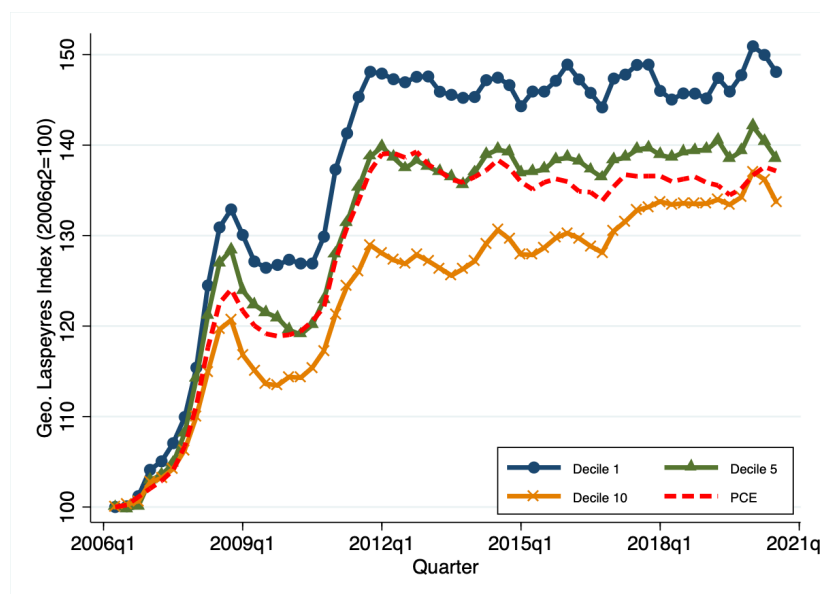


Figure C.9: Price Index for Fats and Oil

Notes: This figure represents the chained geometric Laspeyres price index constructed in the NielsenIQ (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for fats and oil. All else remains the same as before.

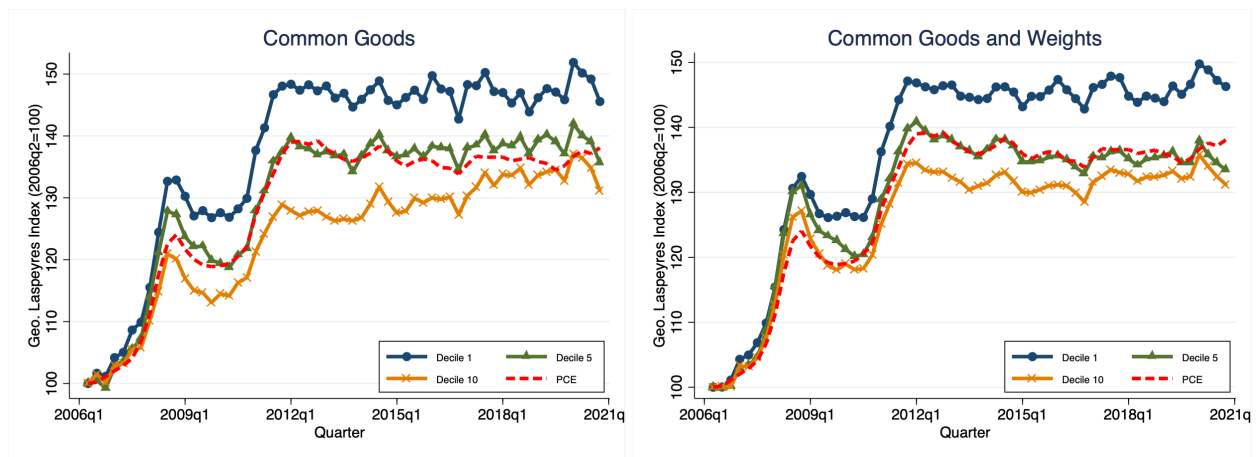


Figure C.10: Price Index for Fats and Oil (Common Goods and Weights)

Notes: This figure represents the chained geometric Laspeyres price index constructed in the NielsenIQ (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for fats and oil. All descriptions remain the same as before except that the solid lines constructed in NielsenIQ are restricted to the set of goods present across all ten deciles in quarters $t - 1$ and t (named “common goods”) in the left panel, and are restricted to these common goods and further based on the same sales weights in decile 1 in the right panel.

D Robustness: Retailer Market Structure

D.1 Other Measures and Controls

We assess the robustness of the regression results in Table 4 by using alternative measures of market concentration (CR1 and CR3) and controlling for MSA-level population. The corresponding results are reported in Tables D.1 and D.2, respectively, which are consistent with our findings.

Table D.1: Retailer Market Structure across Regions with Different Income Levels (CRs)

	CR1	CR3
Income	-0.004*** (0.001)	-0.003*** (0.001)
Quarter FE	Yes	Yes
Observations	11,100	11,100

Note: The table presents regression results from equation (2) by replacing the main dependent variable with the sales share of the top one and three retailers in an MSA for a given quarter in Columns 1 and 2, respectively. The coefficient of interest is on income per capita (in \$1000) in an MSA. All else remains the same as in Table 4. Standard errors are clustered at the MSA level. *** p<0.01, ** p<0.05, * p<0.1

Table D.2: Retailer Market Structure across Regions with Different Income Levels (Population Control)

	Sales (in \$1mil.)	Chain#	Store#	Large Firm% (Store)	HHI
Income	8.482*** (2.979)	0.103*** (0.027)	2.651** (1.059)	-0.007*** (0.002)	-0.003* (0.001)
Population	13.27*** (2.034)	0.066*** (0.009)	10.22*** (0.768)	-0.001*** (0.000)	-0.001*** (0.000)
Quarter FE	Yes	Yes	Yes	Yes	Yes
Observations	11,100	11,100	11,100	11,100	11,100

Note: The table presents regression results from equation (2) by controlling for MSA-level population additionally. The coefficient of interest is on income per capita (in \$1000) in an MSA. All else remains the same as in Table 4. Standard errors are clustered at the MSA level. *** p<0.01, ** p<0.05, * p<0.1

D.2 Long-run Relationship

We estimate the following regression relating income levels to retailer market structure in the long run:

$$\bar{Y}_m = \beta_0 + \beta_1 \bar{Income}_m + \varepsilon_m, \quad (D.1)$$

where $\bar{Y}_m$ is the long-run average of sales, total count of chains or stores, the share of large retailers (defined as the top decile of total sales or the number of stores at the national level), or market concentration (HHI), and $\bar{Income}_m$ is the long-run average per capita income in MSA m . The results are shown in Table D.3, which support the robustness of the main findings. Furthermore, we observe consistent patterns with CR measures, presented in Table D.4.

Table D.3: Retailer Market Structure across Regions with Different Income Levels (Long Run)

	Sales (in \$1mil.)	Chain#	Store#	Large Firm% (Sales)	Large Firm% (Store)	HHI
Income	27.32*** (2.921)	0.198*** (0.023)	17.48*** (2.135)	-0.003*** (0.008)	-0.009*** (0.001)	-0.005*** (0.002)
Observations	185	185	185	185	185	185

Note: The table presents regression results from equation (D.1). The coefficient of interest is on the long-run average of income per capita (in \$1000) in an MSA. The variable definitions remain the same as in Table 4, but based on long-run averages. Standard errors are heteroskedasticity robust. *** p<0.01, ** p<0.05, * p<0.1

Table D.4: Retailer Market Structure across Regions with Different Income Levels (Long Run, CRs)

	CR1	CR3
Income	-0.004*** (0.001)	-0.003*** (0.001)
Observations	185	185

Note: The table presents regression results from equation (D.1). The coefficient of interest is on the long-run average of income per capita (in \$1000) in an MSA. The variable definitions remain the same as in Table D.1, but based on long-run averages. Standard errors are heteroskedasticity robust. *** p<0.01, ** p<0.05, * p<0.1

D.3 Business Dynamics Statistics (BDS)

The Business Dynamics Statistics (BDS) is a public version of administrative Census firm-level data. The data provide annual measures of business dynamics in the U.S., such as job creation and destruction, establishment births and deaths, and firm entry and exit. These data are provided for the economy overall as well as for aggregates defined by establishment or firm characteristics such as firm size and age. Furthermore, the data provide sectoral- and geographic-level information, which allows us to track business dynamics at the sector, state, county, and MSA levels.⁴

In BDS, we focus on the retail trade sector (NAICS 44–45) and measure retailer size by employment.⁵ For each MSA, the Census Bureau reports the number of firms, establishments, total employment, and job creation and destruction, categorized by three firm-size bins: (i) 1–19 employees, (ii) 20–499 employees, and (iii) 500 or more employees. We define large retailers as those in the third category and, for each MSA, count the number of firms and establishments associated with them. To be consistent with NielsenIQ, we focus on the period of 2006–2020 in BDS.⁶

We estimate the following regression to examine cross-sectional patterns in BDS:

$$Y_{mt} = \beta_0 + \beta_1 Income_{mt} + \delta_t + \varepsilon_{mt}, \quad (D.2)$$

where Y_{mt} is the number of firms, establishments, total employment (in thousands), the share of large retailers, and the share of establishments owned by large retailers in MSA m in year t . As before, $Income_{mt}$ represents the income per capita in MSA m in year t , and δ_t denotes a year fixed effect. The results are presented in Table D.5 and are consistent with the baseline findings from NielsenIQ. They also confirm that poorer areas tend to have fewer firms and establishments, as well as a higher share of large firms and establishments.

As in the previous analysis, we examine the long-run cross-sectional relationship across MSAs

⁴See more details at <https://www.census.gov/programs-surveys/bds.html>.

⁵Note that NAICS code 44–45 is more aggregated than ideal for our purposes, but it is the most disaggregated level available in BDS.

⁶In the BDS, 381 MSAs are available based on Census CBSA codes, which we use for our baseline results. The results are robust when restricting the sample to the 185 MSAs covered by NielsenIQ.

Table D.5: Retailer Market Structure (BDS)

	Firm Counts	Estab Counts	Employment	Large Firm Share	Large Estab Share
Income	128.02*** (45.02)	188.96*** (62.18)	3.002*** (0.891)	-0.003*** (0.001)	-0.001*** (0.000)
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	5,715	5,715	5,715	5,715	5,715

Notes: The table represents regression results from equation (D.2). The coefficient of interest is the coefficient on income per capita (in \$1000) in an MSA for a given year. The dependent variable is the total number of firms in Column 1, total number of establishments in Column 2, total employment size (in thousands) in Column 3, the unweighted share of large firms in Column 4, and the unweighted share of establishments associated with large firms in Column 5 in an MSA for a given year. Large firms are defined by those with 500 or more employees. Data is collected from the Business Dynamics Statistics and retailers are gathered from retail trade sector (NAICS 44-45) for 2006-2020. Standard errors are clustered at the MSA level. *** p<0.01, ** p<0.05, * p<0.1

Table D.6: Retailer Market Structure (BDS, Long Run)

	Firm Counts	Estab Counts	Employment	Large Firm Share	Large Estab Share
Income	137.92*** (20.71)	203.61*** (29.15)	3.210*** (0.427)	-0.003*** (0.000)	-0.001*** (0.000)
Observations	381	381	381	381	381

Notes: The table represents regression results from equation (D.3). The variable definitions remain the same as in Table D.5, but based on long-run averages over 2006-2020 in the BDS. Standard errors are heteroskedasticity robust. *** p<0.01, ** p<0.05, * p<0.1

using the following regression:

$$\bar{Y}_m = \beta_0 + \beta_1 \bar{Income}_m + \varepsilon_m, \quad (D.3)$$

where $\bar{Y}_m$ is the long-run average of the number of firms, establishments, total employment (in thousands), the share of large retailers, and the share of establishments owned by large retailers, and $\bar{Income}_m$ is the long-run average of income per capita in MSA m . Table D.6 shows that the BDS data yield patterns consistent with our previous findings.

E MSA-level Coverage: CBP vs. NielsenIQ

One potential concern is that the measured market concentration in NielsenIQ may be affected by differential retailer coverage across MSAs. For example, if NielsenIQ retailer coverage is

systematically more comprehensive in higher-income MSAs, differences in measured market concentration across regions could partly reflect sampling rather than underlying market structure.

To address this concern, we compare NielsenIQ retailer coverage with County Business Patterns (CBP) data from the U.S. Census Bureau and examine whether NielsenIQ coverage varies systematically with MSA income. In the CBP, we use establishments as the unit of analysis, while in NielsenIQ, we use stores as the baseline retail unit and chains as a robustness measure. Throughout this exercise, we use the term “retailers” to refer interchangeably to the relevant unit of analysis in each dataset. We restrict the CBP sample to the Food and Beverage Stores subsector (NAICS 445) and compare it with retailers in the aggregate food and beverages market in NielsenIQ over 2006-2020. As a robustness check, we further restrict the CBP sample to the “Supermarkets and Other Grocery Retailers” industry (NAICS 445110).

First, using CBP, we calculate the share of retailers located in the 185 MSAs covered by NielsenIQ. These MSAs account for 82.13% of all retailers and 80.80% of total employment in the Food and Beverage Stores sector in CBP on average. Second, among these 185 MSAs, we calculate each MSA’s share of total retailers across the 185 MSAs separately in NielsenIQ and CBP, and then construct the ratio of its NielsenIQ retailer share to its corresponding CBP retailer share.⁷ Next, we regress this ratio on MSA-level income per capita to examine whether NielsenIQ is systematically more representative in higher-income MSAs. We estimate the following regression:

$$\left(\frac{RetailerShare_{mt}^{NielsenIQ}}{RetailerShare_{mt}^{CBP}} \right) = \beta IncomePerCapita_{mt} + \delta_m + \delta_t + \varepsilon_{mt}, \quad (E.1)$$

where $\left(\frac{RetailerShare_{mt}^{NielsenIQ}}{RetailerShare_{mt}^{CBP}} \right)$ denotes the relative coverage of retailers in NielsenIQ compared with CBP in MSA m and year t , $IncomePerCapita_{mt}$ denotes MSA-level income per capita (in thousands of dollars), and δ_m and δ_t denote MSA and year fixed effects, respectively.

Tables E.1 and E.2 report the results, with the former based on NAICS 445 and the latter restricted to NAICS 445110. In each table, we consider several specifications. The first three columns measure NielsenIQ coverage using store shares, while the last three columns use retail chain shares, with alternative combinations of year and MSA fixed effects. Across all specifications,

⁷In NielsenIQ, we compute this measure at the quarterly level and then take the average across quarters within each year.

Table E.1: NielsenIQ Retailer Coverage Relative to CBP (NAICS 445)

NielsenIQ/CBP retailer share						
Income	-0.016*	-0.033**	-0.016*	-0.084***	-0.136***	0.006
	(0.009)	(0.014)	(0.009)	(0.027)	(0.042)	(0.029)
Year FE	No	Yes	Yes	No	Yes	Yes
MSA FE	No	No	Yes	No	No	Yes
Observations	2,742	2,742	2,742	2,742	2,742	2,742

Note: The table reports MSA-level regressions of NielsenIQ retailer coverage relative to County Business Patterns (CBP). The dependent variable is the ratio of the MSA-level share of retailers observed in NielsenIQ to that in CBP. In CBP, we restrict the sample to the Food and Beverage Stores subsector (NAICS 445). The independent variable is MSA-level income per capita (in thousands of dollars). Columns 1,2, and 3 use the share of stores in an MSA, and Columns 4,5, and 6 use the share of retail chains in an MSA in NielsenIQ. Columns 1 and 4 include no fixed effects, Columns 2 and 5 include year fixed effects only, and Columns 3 and 6 include year and MSA fixed effects. Standard errors are clustered at the MSA level. *** p<0.01, ** p<0.05, * p<0.1.

Table E.2: NielsenIQ Retailer Coverage Relative to CBP (NAICS 445110)

NielsenIQ/CBP retailer share						
Income	-0.014	-0.031**	-0.026**	-0.069**	-0.115**	-0.003
	(0.009)	(0.013)	(0.011)	(0.034)	(0.051)	(0.046)
Year FE	No	Yes	Yes	No	Yes	Yes
MSA FE	No	No	Yes	No	No	Yes
Observations	2,742	2,742	2,742	2,742	2,742	2,742

Note: The table reports MSA-level regressions of NielsenIQ retailer coverage relative to County Business Patterns (CBP). The dependent variable is the ratio of the MSA-level share of retailers observed in NielsenIQ to that in CBP. In CBP, we restrict the sample to the Supermarkets and Other Grocery Stores subsector (NAICS 445110). The independent variable is MSA-level income per capita (in thousands of dollars). Columns 1,2, and 3 use the share of stores in an MSA, and Columns 4,5, and 6 use the share of retail chains in an MSA in NielsenIQ. Columns 1 and 4 include no fixed effects, Columns 2 and 5 include year fixed effects only, and Columns 3 and 6 include year and MSA fixed effects. Standard errors are clustered at the MSA level. *** p<0.01, ** p<0.05, * p<0.1.

NielsenIQ coverage is not relatively higher in higher-income MSAs compared with CBP.⁸ These results alleviate the concern that the higher retailer concentration we measure in lower-income MSAs is mechanically driven by NielsenIQ providing systematically better coverage in richer areas.

⁸If anything, when exploiting cross-sectional variation across MSAs without MSA fixed effects, higher-income MSAs exhibit relatively lower coverage in NielsenIQ.

Table F.1: Food Inflation across Regions (CRs)

	Inflation	Inflation	Inflation	Inflation
CR1	0.297*** (0.040)	0.286*** (0.042)	0.278*** (0.041)	0.299*** (0.042)
Income		-0.002* (0.001)	-0.001 (0.001)	-0.002** (0.001)
Population			-0.001 (0.001)	-0.002*** (0.001)
Chain #				0.014*** (0.003)
	Inflation	Inflation	Inflation	Inflation
CR3	0.453*** (0.063)	0.432*** (0.068)	0.414*** (0.068)	0.473*** (0.065)
Income		-0.002 (0.001)	-0.001 (0.001)	-0.002** (0.001)
Population			-0.001 (0.001)	-0.001** (0.001)
Chain #				0.015*** (0.003)
Quarter FE	Yes	Yes	Yes	Yes
Observations	10,730	10,730	10,730	10,730

Note: The table presents regression results from equation (3). The coefficient of interest is on the CR measure (CR1 in the top panel and CR3 in the bottom panel) in an MSA for a given quarter. The dependent variable is the geometric Laspeyres inflation rate (%) of aggregate food in an MSA for a given quarter. MSA-level income per capita (in \$1,000s) is included as a control starting in the second column, MSA population (in 100,000s), measured annually using BEA data, is included as an additional control in the last two columns. In addition, the total number of chains is included in the final column. Quarter fixed effects are included in all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

F Robustness: Inflation and Market Concentration

F.1 Other Measures and Controls

In the main regression (3), we replace HHI with CR1 and CR3 measures for market concentration and report the results in Table F.1. The results are robust to both measures.

Table F.2: Food Inflation across Regions (Long Run)

	Inflation	Inflation	Inflation	Inflation
HHI	0.267*** (0.046)	0.244*** (0.049)	0.231*** (0.048)	0.275*** (0.052)
Income		-0.002* (0.001)	-0.001 (0.001)	-0.003** (0.001)
Population			-0.001 (0.001)	-0.002*** (0.001)
Chain #				0.017*** (0.004)
Observations	185	185	185	185

Note: The table presents regression results from equation (F.1) for the cross-sectional association between MSA-level inflation and market concentration in the long run. Each variable is measured as its long-run average across quarters within an MSA, leaving one observation per MSA. The dependent variable is the long-run average of the geometric Laspeyres inflation rate (%) for aggregate food. The coefficients of interest are on the long-run averages of the HHI of retail chain sales in an MSA. MSA-level income per capita (in \$1,000s) is included as a control starting in the second column, MSA population (in 100,000s), measured annually using BEA data, is included as an additional control in the last two columns. In addition, the total number of chains is included in the final column. Standard errors are heteroskedasticity robust. *** p<0.01, ** p<0.05, * p<0.1

F.2 Long-run Relationship

Furthermore, we explore the long-run association between inflation and market concentration across MSAs with the following regression:

$$\bar{P}_m = \beta_0 + \beta_1 \bar{H} \bar{H} I_m + \varepsilon_m, \quad (\text{F.1})$$

where $\bar{P}_m$ is the long-run average of inflation, and $\bar{H} \bar{H} I_m$ is the long-run average of HHI in MSA m . The results are presented in Table F.2, showing a positive association between food inflation and market concentration across MSAs in the long run. Also, the long-run patterns hold for both CR1 and CR3 measures as shown in Table F.3.

F.3 Price Indexes

In the main regression (3), we replace the dependent variable with the geometric Laspeyres price index at the MSA level to capture cumulative price differences across MSAs. The results are

Table F.3: Food Inflation across Regions (Long Run, CRs)

	Inflation	Inflation	Inflation	Inflation
CR1	0.295*** (0.048)	0.276*** (0.052)	0.267*** (0.050)	0.283*** (0.051)
Income		-0.002** (0.001)	-0.001 (0.001)	-0.003** (0.001)
Population			-0.001 (0.001)	-0.002** (0.001)
Chain #				0.015*** (0.004)
CR3	0.535*** (0.088)	0.502*** (0.097)	0.480*** (0.097)	0.528*** (0.094)
Income		-0.002 (0.001)	-0.001 (0.001)	-0.003** (0.001)
Population			-0.000 (0.001)	-0.001** (0.001)
Chain #				0.016*** (0.004)
Observations	185	185	185	185

Note: The table presents regression results from equation (F.1) for the cross-sectional association between MSA-level inflation and market concentration in the long run. Each variable is measured as its long-run average across quarters within an MSA, leaving one observation per MSA. The dependent variable is the long-run average of the geometric Laspeyres inflation rate (%) for aggregate food. The coefficient of interest is on the CR measure (CR1 in the top panel and CR3 in the bottom panel) in an MSA. MSA-level income per capita (in \$1,000s) is included as a control starting in the second column, MSA population (in 100,000s), measured annually using BEA data, is included as an additional control in the last two columns. In addition, the total number of chains is included in the final column. Standard errors are heteroskedasticity robust. *** p<0.01, ** p<0.05, * p<0.1

similar and reflected in Table F.4. Furthermore, we replace HHI with CR1 and CR3 for market concentration, and the results remain robust, as presented in Table F.5.

F.4 Eggs Inflation

We estimate the following OLS regression, focusing specifically on egg prices:

$$P_{mt} = \beta_0 + \beta_1 HHI_{mt} + \delta_t + \varepsilon_{mt}, \quad (\text{F.2})$$

Table F.4: Food Price Indexes across Regions

	Price Index	Price Index	Price Index	Price Index
HHI	12.426*** (1.736)	11.409*** (1.809)	10.874*** (1.750)	11.933*** (1.861)
Income		-0.133** (0.054)	-0.084 (0.064)	-0.117** (0.058)
Population			-0.038* (0.022)	-0.059** (0.026)
Chain #				0.342** (0.140)
Quarter FE	Yes	Yes	Yes	Yes
Observations	10,915	10,915	10,915	10,915

Note: The table presents regression results from equation (3) with the dependent variable replaced by the geometric Laspeyres price index. The coefficients of interest are on the HHI of retail chain sales in an MSA for a given quarter. MSA-level income per capita (in \$1,000s) is included as a control starting in the second column, MSA population (in 100,000s), measured annually using BEA data, is included as an additional control in the last two columns. In addition, the total number of chains is included in the final column. Quarter fixed effects are included in all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

where P_{mt} is the geometric Laspeyres inflation rate of eggs, HHI_{mt} denotes the egg-sales-based retailer HHI in MSA m in quarter t , and δ_t represents quarter fixed effects.

The results are presented in Table F.6, which reveal a similar association between HHI and egg inflation to that found for aggregate food in our main analysis. In particular, we find a positive and statistically significant relationship between market concentration and inflation in Column 1, and this relationship remains robust even after controlling for the MSA-level income per capita, population, and the number of chains in Columns 2, 3, and 4. We also replace HHI with CR1 and CR3 as alternative measures of market concentration in the egg market. Table F.7 shows that the positive association between market concentration and egg inflation is robust to both alternative measures.

While these results shed light on the link between inflation and market concentration, they do not establish causality. The OLS estimate of β_1 may be subject to endogeneity bias. For instance, the observed relationship could be demand-driven: consumers in MSAs with higher HHI may disproportionately purchase goods experiencing higher inflation. Alternatively, income-related differences in consumer behavior could play a role: richer MSAs may have consumers who are more sensitive to price changes. Such heterogeneity in consumer behavior may have led retailers

Table F.5: Food Price Indices across Regions (CRs)

	Price Index	Price Index	Price Index	Price Index
CR1	11.888*** (1.743)	11.032*** (1.824)	10.641*** (1.753)	11.020*** (1.785)
Income		-0.141*** (0.053)	-0.086 (0.063)	-0.111* (0.058)
Population			-0.041* (0.022)	-0.058** (0.026)
Chain #				0.251* (0.134)
	Price Index	Price Index	Price Index	Price Index
CR3	16.098*** (2.784)	14.372*** (2.831)	13.212*** (2.745)	14.260*** (2.724)
Income		-0.136** (0.053)	-0.094 (0.063)	-0.121** (0.058)
Population			-0.034 (0.022)	-0.051** (0.026)
Chain #				0.275** (0.138)
Quarter FE	Yes	Yes	Yes	Yes
Observations	10,915	10,915	10,915	10,915

Note: The table presents regression results from equation (3) with the dependent variable replaced by the geometric Laspeyres price index. The coefficients of interest are on the CR measure (CR1 in the top panel and CR3 in the bottom panel) in an MSA for a given quarter. MSA-level income per capita (in \$1,000s) is included as a control starting in the second column, MSA population (in 100,000s), measured annually using BEA data, is included as an additional control in the last two columns. In addition, the total number of chains is included in the final column. Quarter fixed effects are included in all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

in wealthier areas to raise prices at slower rates. Another potential explanation is a supply-side story, where poorer MSAs have fewer stores and varieties, which weakens competition and allows retailers to raise prices more aggressively. There may also be a potential sorting of certain types of retailers into poorer MSAs, those that are more flexible in increasing prices, compared to retailers operating in richer areas. To determine whether the effect we observe is driven by supply-side or demand-side forces, we exploit the 2014–2015 bird flu outbreak as an exogenous supply shock in Section 4 of the main text.

Table F.6: Eggs Inflation across Regions

	Inflation	Inflation	Inflation	Inflation
HHI	0.514*** (0.150)	0.440*** (0.159)	0.411** (0.162)	0.483*** (0.184)
Income		-0.011*** (0.004)	-0.008* (0.004)	-0.010** (0.005)
Population			-0.002** (0.001)	-0.003** (0.001)
Chain #				0.023 (0.020)
Quarter FE	Yes	Yes	Yes	Yes
Observations	8,743	8,743	8,743	8,743

Note: The table represents regression results from (F.2) for the cross-sectional association between MSA-level inflation and market concentration for eggs. The coefficient of interest is on HHI of retail chain's egg sales in an MSA for a given quarter. The dependent variable is the geometric Laspeyres inflation rate (%) of eggs in an MSA for a given quarter. MSA-level income per capita (in \$1,000s) is included as a control starting in the second column, MSA population (in 100,000s), measured annually using BEA data, is included as an additional control in the last two columns. In addition, the total number of chains is included in the final column. Quarter fixed effects are included in all specifications. Standard errors are clustered at the MSA level. *** p<0.01, ** p<0.05, * p<0.1

Table F.7: Eggs Inflation across Regions (CRs)

	Inflation	Inflation	Inflation	Inflation
CR1	0.633*** (0.180)	0.571*** (0.190)	0.545*** (0.193)	0.587*** (0.206)
Income		-0.012*** (0.004)	-0.008* (0.005)	-0.010** (0.005)
Population			-0.002** (0.001)	-0.003** (0.001)
Chain #				0.019 (0.019)
	Inflation	Inflation	Inflation	Inflation
CR3	1.083*** (0.353)	0.956** (0.385)	0.894** (0.413)	0.918** (0.415)
Income		-0.011** (0.004)	-0.009* (0.005)	-0.010** (0.005)
Population			-0.002 (0.001)	-0.002* (0.001)
Chain #				0.012 (0.017)
Quarter FE	Yes	Yes	Yes	Yes
Observations	8,743	8,743	8,743	8,743

Note: The table represents regression results from equation (F.2) by replacing HHI with CR1 (in the top panel) and CR3 (in the bottom panel) for eggs. The coefficient of interest is on CR1 and CR3 of retail chain's eggs sales in an MSA. The dependent variable is the geometric Laspeyres inflation rate (%) of eggs in an MSA for a given quarter. MSA-level income per capita (in \$1,000s) is included as a control starting in the second column, MSA population (in 100,000s), measured annually using BEA data, is included as an additional control in the last two columns. In addition, the total number of chains is included in the final column. Quarter fixed effects are included in all specifications. Standard errors are clustered at the MSA level. *** p<0.01, ** p<0.05, * p<0.1

G Egg Market

Note that the egg market in general tends to be local and regional. For instance, Cal-Maine Foods, which is the largest producer and marketer of eggs in the U.S., primarily operates in regional markets. See Figure G.1 that plots the locations of Cal-Maine Foods, which are mainly concentrated in Southern areas.

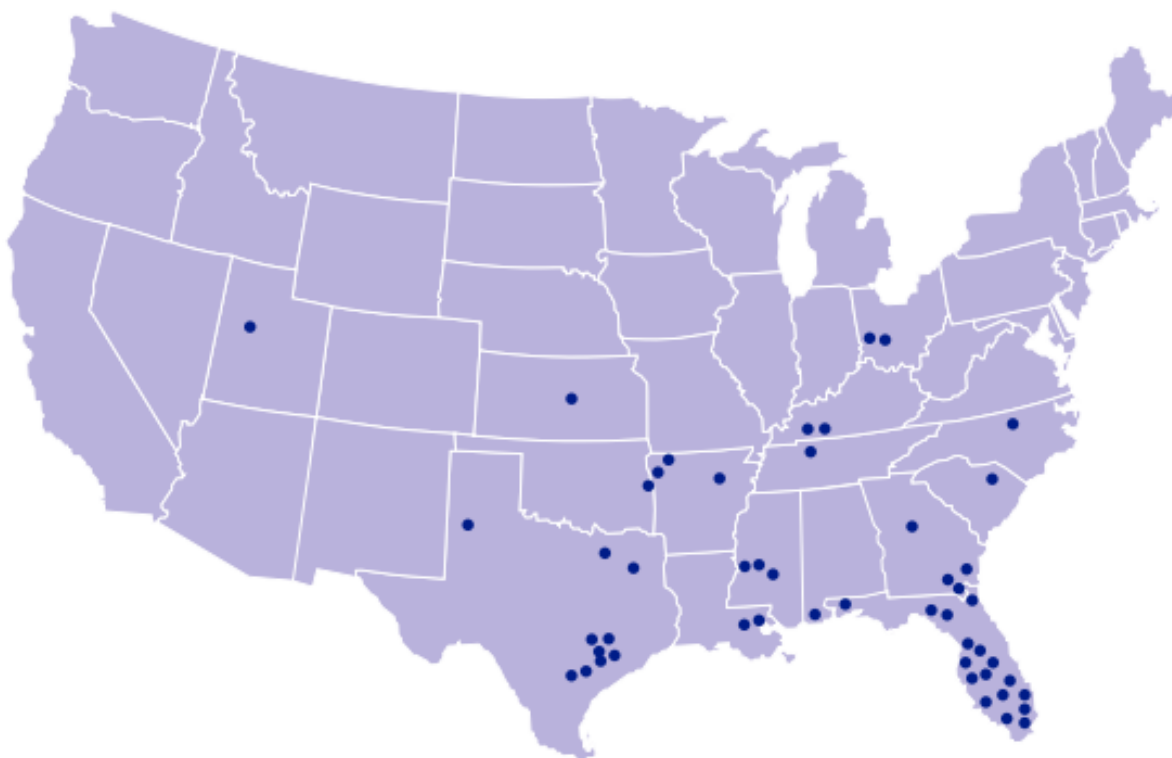


Figure G.1: Location of Cal-Maine Foods

Notes: The figure represents the local locations of Cal-Maine Foods, which is the largest national egg-producing company in the United States.

H Robustness: Triple-Difference Estimation

H.1 Terciles of HHI

We use HHI terciles to assess the robustness of our main triple-difference results. Specifically, we estimate the following regression:

$$P_{mt} = \beta_0 + \beta_1(Post_t \times HighHHI_m) + \beta_2(Post_t \times MidHHI_m)$$

$$\begin{aligned}
& + \beta_3(BirdFlu_m \times Post_t) + \beta_4(BirdFlu_m \times Post_t \times MidHHI_m) \\
& + \beta_5(BirdFlu_m \times Post_t \times HighHHI_m) + \delta_m + \delta_t + \varepsilon_{mt},
\end{aligned} \tag{H.1}$$

where subscript m indexes MSAs and t indexes quarters. $BirdFlu_m$ and $Post_t$ are defined the same as before. We classify MSAs into terciles based on the distribution of egg-market sales concentration (HHI) in 2014Q3. The bottom tercile ($LowHHI$) comprises the one-third of MSAs with the lowest HHI and serves as the reference group.⁹ $HighHHI_m$ is an indicator for the MSAs in the top tercile, and $MidHHI_m$ is an indicator for those in the middle tercile. P_{mt} denotes the geometric Laspeyres quarterly inflation rate for eggs in MSA m in quarter t . MSA and quarter fixed effects, δ_m and δ_t , are included as before, and ε_{mt} is the error term.

The results are presented in Table H.1. Column 1 pools all quarters within the four-year window and shows that, among treated MSAs, those in the top tercile of retail market concentration experience 2.3 percentage points higher inflation, on average, than those in the bottom tercile. This effect is statistically significant at the 1% level. Columns 2 and 3 decompose this effect into the inflationary and deflationary periods, respectively. Restricting the sample to the inflationary period, Column 2 shows that treated MSAs in the top tercile exhibit 4.7 percentage points higher inflation, on average, than treated MSAs in the bottom tercile, significant at the 1% level. In contrast, Column 3 shows no evidence that highly concentrated treated MSAs experience larger price declines during the deflationary period. The estimated effect is positive but statistically insignificant, suggesting persistence in the initial inflationary response. These results are consistent with our main findings.

Furthermore, we use HHI terciles to assess the robustness of our UPC-MSA-quarter-level results. We estimate the following regression, where the dependent variable is the log change in UPC-level prices within the egg market:

$$\begin{aligned}
\Delta \ln price_{umt} = & \beta_0 + \beta_1(Post_t \times HighHHI_m) + \beta_2(Post_t \times MidHHI_m) \\
& + \beta_3(BirdFlu_m \times Post_t) + \beta_4(BirdFlu_m \times Post_t \times MidHHI_m) \\
& + \beta_5(BirdFlu_m \times Post_t \times HighHHI_m) + \delta_u + \delta_m + \delta_t + \varepsilon_{umt},
\end{aligned} \tag{H.2}$$

⁹We fix HHI to its 2014Q3 value, the quarter preceding the outbreak, to mitigate endogeneity concerns and isolate the effect of the supply shock on price dynamics. Results are robust to fixing HHI to its two-year average over 2012Q4–2014Q3.

Table H.1: Triple Difference Estimator (Eggs)

	Inflation	Inflation	Inflation
Bird Flu $\times$ Post $\times$ HighHHI	0.023*** (0.006)	0.047*** (0.014)	0.015 (0.012)
Bird Flu $\times$ Post $\times$ MidHHI	0.003 (0.009)	0.023** (0.010)	-0.005 (0.016)
Bird Flu $\times$ Post	-0.012** (0.005)	0.016** (0.008)	-0.038*** (0.005)
Post $\times$ HighHHI	-0.003 (0.002)	-0.002 (0.004)	-0.002 (0.006)
Post $\times$ MidHHI	0.000 (0.002)	0.000 (0.004)	0.002 (0.006)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Observations	3,145	1,850	1,295

Note: The table represents regression results from equation (H.1). The coefficient of interest is the interaction of Bird Flu, Post, and High HHI. Bird Flu is a binary variable that takes the value of one for MSAs in which egg farmers culled their layers during the 2014-2015 bird flu episode, and Post is a binary variable that takes the value of one in the post-shock period after 2014Q4. High HHI is an indicator for the top tercile MSAs, and Mid HHI is an indicator for the mid tercile MSAs in the distribution of sales HHI in 2014Q3. The sample period ranges from 2012Q4 to 2016Q4. Inflationary and deflationary periods are determined by the national price index of eggs. Column 1 pools all periods together, Column 2 only considers the inflationary period, and Column 3 only considers the deflationary period. MSA and quarter fixed effects are included across all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

where $\Delta \ln price_{umt}$ denotes the log difference in the price of UPC u in MSA m between quarters $t - 1$ and t . The UPC fixed effects, δ_u , account for regional variation in consumption baskets by ensuring that price changes are identified within identical products. All other terms are defined as in equation (H.1).

Table H.2 reports the UPC-level results. In Column 1, pooling all quarters, we find that following the bird flu shock, egg prices in exposed MSAs in the highest concentration tercile increased by an additional 1 percentage point relative to exposed MSAs in the bottom tercile within the same UPCs. This estimate is statistically significant at the 1 percent level. Restricting the sample to the inflationary period, Column 2 shows that the triple interaction for the highest HHI tercile remains positive and significant at the 1 percent level. Quantitatively, prices rose by an additional 2 percentage points in highly concentrated exposed MSAs relative to less concentrated exposed

Table H.2: Triple Difference Estimator (Eggs, UPC-level)

	$\Delta \ln \text{Price}$	$\Delta \ln \text{Price}$	$\Delta \ln \text{Price}$
Bird Flu $\times$ Post $\times$ HighHHI	0.010*** (0.004)	0.020*** (0.006)	-0.007 (0.008)
Bird Flu $\times$ Post $\times$ MidHHI	0.005 (0.004)	0.019*** (0.005)	-0.012 (0.008)
Bird Flu $\times$ Post	-0.006* (0.003)	0.000 (0.005)	-0.013* (0.007)
Post $\times$ HighHHI	-0.004*** (0.002)	-0.005 (0.003)	-0.002 (0.003)
Post $\times$ MidHHI	-0.001 (0.002)	-0.003 (0.003)	0.002 (0.004)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
UPC FE	Yes	Yes	Yes
Observations	145,989	84,418	61,470

Note: The table represents regression results from equation (H.2). The dependent variable is the log price difference of product u in MSA m between quarters $t - 1$ and t . All descriptions remain the same as before, except that MSA, UPC, and quarter fixed effects are included across all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

MSAs for identical egg products. In Column 3, focusing on the deflationary period, the estimated effect is statistically insignificant, suggesting persistence in the inflationary response rather than symmetric price reversals.

H.2 CR Measures

To test the robustness of the triple-difference regression in equation (5), we replace HHI with CR1 and CR3 in the egg market. The results are robust, as displayed in Table H.3.

H.3 Neighboring Treated MSAs

We incorporate five additional neighboring MSAs around the affected areas using MSA maps provided by the Census Bureau.¹⁰ These MSAs are listed in Table H.4. We then re-estimate regressions based on equation (4) and equation (5) including these additional treated MSAs. The results, presented in Tables H.5 and H.6, are consistent with our baseline findings.

¹⁰See details at <https://www.census.gov/geographies/reference-maps/2020/demo/state-maps.html>.

Table H.3: Triple Difference Estimator (CRs)

	Inflation	Inflation	Inflation
Bird Flu $\times$ Post $\times$ CR1	0.052*** (0.010)	0.093*** (0.022)	0.044* (0.024)
Bird Flu $\times$ Post	-0.037*** (0.008)	-0.021 (0.013)	-0.064*** (0.015)
CR1 $\times$ Post	-0.010* (0.005)	-0.008 (0.008)	-0.006 (0.013)
Bird Flu $\times$ CR1	-0.139 (0.090)	-0.204* (0.110)	-0.092 (0.056)
CR1	0.046** (0.022)	0.060** (0.027)	0.045 (0.028)
	Inflation	Inflation	Inflation
Bird Flu $\times$ Post $\times$ CR3	0.098*** (0.030)	0.191*** (0.063)	0.099* (0.055)
Bird Flu $\times$ Post	-0.092*** (0.028)	-0.134** (0.057)	-0.125*** (0.048)
CR3 $\times$ Post	-0.023* (0.012)	-0.021 (0.021)	-0.029 (0.030)
Bird Flu $\times$ CR3	0.003 (0.120)	0.037 (0.232)	-0.194 (0.150)
CR3	0.049 (0.043)	0.038 (0.057)	0.071 (0.088)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Observations	3,145	1,850	1,295

Note: The table represents regression results from equation (5) by replacing HHI with CR1 or CR3. The coefficient of interest is the interaction of Bird Flu, Post, and CR1 (in the top panel) or CR3 (in the bottom panel). CR1 (CR3) is the concentration ratio of the top 1 (top 3) retail chain's sales of eggs within an MSA, of which value is fixed to 2014Q3 for all quarters in the post period. All other descriptions remain the same as in Table 7. Standard errors are clustered at the MSA-level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

H.4 Placebo Test

One limitation of our identification strategy is that the triple-difference design identifies whether MSAs with higher market concentration experienced larger inflation responses to the bird flu shock, but it does not by itself prove that market concentration is the underlying mechanism. In other words, the estimates show that inflation rose more in bird-flu-exposed, high-HHI MSAs than in bird-flu-exposed, low-HHI MSAs, relative to unaffected MSAs. However, high-HHI MSAs may

Table H.4: Neighboring MSAs around the Impacted MSAs

#	MSA	State
1	CEDAR RAPIDS-WATERLOO & DUBUQUE	IA
2	MILWAUKEE	WI
3	LA CROSSE-EAU CLAIRE	WI
4	LINCOLN & HASTINGS-KEARNY	NE
5	DULUTH-SUPERIOR	MN-WI

Note: The table provides the set of neighboring MSAs around the impacted MSAs.

Table H.5: Difference-in-Differences Estimator (Neighbor MSAs)

	Inflation	Inflation	Inflation	Abs. Inflation
Bird Flu $\times$ Post	-0.003 (0.004)	0.037*** (0.006)	-0.033*** (0.007)	0.047*** (0.005)
Sample Periods	All	Inflation	Deflation	All
Quarter FE	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes
Observations	3,145	1,850	1,295	3,145

Note: The table represents regression results from equation (4), where the treated MSAs include five additional neighboring MSAs around the impacted areas. All other descriptions remain the same as in the main Table 6. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

differ from low-HHI MSAs along other dimensions that also affect pass-through from cost shocks to prices. If those characteristics are correlated with HHI, the triple interaction could partly reflect these other local differences rather than market concentration itself.

To assess this concern, we conduct a placebo test using income per capita, a local characteristic that is correlated with market concentration, as shown in Section 3. Specifically, we replace HHI with income per capita and estimate the same triple-difference specification in equation (5). Table H.7 shows that the placebo results differ from the baseline HHI results in three ways. First, the triple-interaction coefficient is statistically insignificant in two of the three specifications. Second, in the inflationary-period specification (column 2), the coefficient is negative and only marginally significant at the 10 percent level. Third, the sign and pattern of the estimates do not resemble the baseline results, where higher HHI consistently predicts stronger inflation responses to the bird flu shock. These results suggest that the baseline findings are unlikely to be driven simply by broad differences between higher- and lower-income MSAs, although they do not rule out all possible correlated local mechanisms.

Table H.6: Triple Difference Estimator (Neighbor MSAs)

	Inflation	Inflation	Inflation
Bird Flu $\times$ Post $\times$ HHI	0.049*** (0.008)	0.074*** (0.019)	0.056** (0.024)
Bird Flu $\times$ Post	-0.028*** (0.007)	-0.001 (0.010)	-0.062*** (0.012)
HHI $\times$ Post	-0.011** (0.005)	-0.008 (0.008)	-0.010 (0.011)
Bird Flu $\times$ HHI	-0.129 (0.087)	-0.198 (0.134)	-0.002 (0.072)
HHI	0.039 (0.025)	0.054* (0.030)	0.031 (0.041)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Observations	3,145	1,850	1,295

Note: The table represents regression results from equation (5), where the treated MSAs include five additional neighboring MSAs around the impacted areas. All other descriptions remain the same as in Table 7. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

H.5 Additional Income and Price Level Controls

Section H.4 shows that income per capita does not drive our main triple-difference results. A remaining concern, however, is that the estimated effect of market concentration may be sensitive to controlling more flexibly for income differences across MSAs. As an additional robustness check, we augment the baseline specification by adding the triple interaction of Bird Flu, Post, and income per capita, along with the corresponding lower-order interaction terms. Table H.8 reports the results.

The coefficient on Bird Flu $\times$ Post $\times$ HHI remains positive in all three specifications. It is statistically significant at the 1 percent level in the pooled sample and during the inflationary period, with estimates of 0.084 and 0.110, respectively. In the deflationary period, the coefficient is smaller and statistically insignificant. This positive estimate is nevertheless consistent with the baseline results, where retailers in more concentrated markets reduce prices more slowly when aggregate prices are falling. Moreover, the coefficient on Bird Flu $\times$ Post $\times$ income per capita is statistically insignificant in two of the three specifications and only marginally significant in the pooled sample. Taken together, these results further indicate that the main triple-difference findings are not driven by income differences across MSAs.

Table H.7: Placebo Triple Difference Estimator (Eggs, Income)

	Inflation	Inflation	Inflation
Bird Flu $\times$ Post $\times$ Income	-0.006 (0.011)	-0.031* (0.018)	0.001 (0.011)
Bird Flu $\times$ Post	0.066 (0.060)	0.232** (0.096)	-0.066 (0.052)
Income $\times$ Post	0.002* (0.001)	-0.001 (0.003)	0.006* (0.003)
Bird Flu $\times$ Income	-0.126*** (0.035)	-0.117*** (0.033)	0.057* (0.030)
Income	0.005 (0.009)	-0.055*** (0.013)	0.021* (0.013)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Observations	3,145	1,850	1,295

Note: The table represents regression results from equation (5). The coefficient of interest is the interaction of Bird Flu, Post, and Income. Bird Flu is a binary variable that takes the value of one for MSAs in which egg farmers culled their layers during the 2014-2015 bird flu episode, and Post is a binary variable that takes the value of one in the post-shock period after 2014Q4. Income is income per capita (in \$10,000), which is measured annually at the MSA level from BEA data. The sample period ranges from 2012Q4 to 2016Q4. Inflationary and deflationary periods are determined by the national price index of eggs. Column 1 pools all periods together, Column 2 only considers the inflationary period, and Column 3 only considers the deflationary period. MSA and quarter fixed effects are included across all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

A related concern is that the estimated effect of market concentration may instead reflect differences in price levels across MSAs, rather than market power per se (Sangani, 2026). As an additional robustness check, we augment the baseline specification by adding the triple interaction of Bird Flu, Post, and Initial Price, along with the corresponding lower-order interaction terms. Initial price is defined as the mean price of a product-MSA pairing over the pre-shock quarters, 2012Q4–2014Q3. Table H.9 reports the results.

The coefficient on Bird Flu $\times$ Post $\times$ HHI remains positive and statistically significant at the 1 percent level in the pooled sample and during the inflationary period, with estimates of 0.020 and 0.036, respectively. In the deflationary period, the coefficient is smaller in magnitude, changes sign, and is statistically insignificant. These results are consistent with our baseline results in equation (6): we continue to find higher pass-through of costs in MSAs with higher market

Table H.8: Triple Difference Estimator (Eggs, Income Controls)

	Inflation	Inflation	Inflation
Bird Flu $\times$ Post $\times$ HHI	0.084*** (0.026)	0.110*** (0.027)	0.032 (0.025)
Bird Flu $\times$ Post	-0.107* (0.064)	0.028 (0.075)	-0.181** (0.076)
HHI $\times$ Post	-0.008 (0.005)	-0.015** (0.008)	-0.000 (0.011)
Bird Flu $\times$ HHI	-0.222 (0.189)	-0.222 (0.225)	-0.377*** (0.124)
HHI	0.036 (0.025)	0.061** (0.027)	0.028 (0.046)
Bird Flu $\times$ Post $\times$ Income	0.022* (0.012)	0.000 (0.015)	0.022 (0.014)
Income $\times$ Post	0.002 (0.001)	-0.001 (0.003)	0.006* (0.003)
Bird Flu $\times$ Income	-0.163*** (0.048)	-0.150*** (0.047)	0.038 (0.028)
Income	0.004 (0.009)	-0.057*** (0.013)	0.021* (0.013)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Observations	3,145	1,850	1,295

Note: The table represents regression results from equation (5). The coefficient of interest is the interaction of Bird Flu, Post, and HHI. Bird Flu is a binary variable that takes the value of one for MSAs in which egg farmers culled their layers during the 2014-2015 bird flu episode, and Post is a binary variable that takes the value of one in the post-shock period after 2014Q4. HHI is the Herfindahl-Hirschman index (HHI) of retail chain's sales of eggs within an MSA, fixed to 2014Q3 values for all quarters in the post period. We add additional controls where we replace HHI with Income in the triple interaction and the corresponding interaction terms. Income is income per capita (in \$10,000), which is measured annually at the MSA level from BEA data. The sample period ranges from 2012Q4 to 2016Q4. Inflationary and deflationary periods are determined by the national price index of eggs. Column 1 pools all periods together, Column 2 only considers the inflationary period, and Column 3 only considers the deflationary period. MSA and quarter fixed effects are included across all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

concentration. Moreover, the coefficient on Bird Flu $\times$ Post $\times$ Initial Price is small in magnitude and statistically insignificant in all three specifications. Taken together, these results suggest that our main triple-difference findings are not driven by differences in initial price levels across MSAs.

Table H.9: Triple Difference Estimator (Eggs, UPC Level, and Initial Price Controls)

	$\Delta \ln \text{ Price}$	$\Delta \ln \text{ Price}$	$\Delta \ln \text{ Price}$
Bird Flu $\times$ Post $\times$ HHI	0.020*** (0.006)	0.036*** (0.010)	-0.014 (0.016)
Bird Flu $\times$ Post	-0.012*** (0.004)	-0.007 (0.006)	-0.010 (0.011)
HHI $\times$ Post	-0.010*** (0.003)	-0.014** (0.007)	0.003 (0.007)
Bird Flu $\times$ HHI	-0.014 (0.060)	-0.043 (0.084)	-0.081 (0.081)
HHI	0.028* (0.015)	0.054** (0.024)	-0.021 (0.025)
Bird Flu $\times$ Post $\times$ Initial Price	0.006 (0.006)	0.003 (0.007)	-0.001 (0.019)
Post $\times$ Initial Price	-0.009*** (0.003)	-0.067*** (0.004)	0.106*** (0.009)
Bird Flu $\times$ Initial Price	0.010** (0.005)	0.004 (0.005)	0.027*** (0.007)
Initial Price	0.031*** (0.007)	0.035*** (0.010)	0.005 (0.024)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Observations	133,653	81,687	51,878

Note: The table reports estimates from equation (6) augmented with a second triple interaction, and presents two coefficients of interest. The first triple interaction is the interaction of Bird Flu, Post, and HHI, as in equation (6). The second triple interaction is the interaction of Bird Flu, Post, and Initial Price, which tests whether the bird-flu shock's pass-through into prices depends on the initial, pre-shock price level of a given product-MSA pair. Bird Flu is a binary variable equal to one for MSAs in which egg farmers culled their layers during the 2014–2015 bird flu episode, and Post is a binary variable equal to one in the post-shock period beginning 2014Q4. HHI is the Herfindahl-Hirschman index of retail chains' egg sales within an MSA, fixed at its 2014Q3 value for all subsequent quarters. Initial Price is each product-MSA pair's mean price in the quarters before the bird-flu episode; because it requires pre-shock observations, the sample is restricted to product-MSA pairs present before 2014Q4, a modestly smaller sample than in equation (6). Both triple interactions are estimated jointly, together with their constituent lower-order interactions and level terms. The sample period ranges from 2012Q4 to 2016Q4. Inflationary and deflationary periods are determined by the national price index of eggs. Column 1 pools all periods together, Column 2 restricts to the inflationary period, and Column 3 restricts to the deflationary period. MSA, quarter, and product (UPC) fixed effects are included in all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

H.6 Retailer Composition

An alternative explanation for the observed differences in inflation across MSAs is variation in retailer composition, as retailers may differ in the extent to which they pass through shocks to prices. In particular, pass-through may differ between national, regional, and local chains for several reasons. On the one hand, national chains—with a larger number of stores spread across diverse regions—may be better able to hedge against local shocks and less responsive to them, resulting in lower pass-through in affected areas.¹¹ On the other hand, national chains may charge prices closer to marginal costs, which could impact the local level of prices, markups, and pass-through.¹² Following this logic, their pass-through of cost shocks will be higher as the market will operate closer to a competitive market. In either case, the presence of national versus local retail chains can influence the degree of pass-through in affected areas.¹³ This suggests that variation in the presence of national versus local chains across MSAs may offer an alternative explanation for our results. Under this interpretation, geographic differences in inflation still originate on the retailer side, but are driven by differences in retailer composition rather than market concentration.

To test this, we follow [Jarmin et al. \(2009\)](#) and classify retailers into three categories based on their geographic footprint: (i) local retailers, operating in only one state; (ii) regional retailers, operating in two to ten states; and (iii) national retailers, operating in more than ten states. To account for the possibility that differential pass-through reflects retailer geographic scope rather than market concentration, we augment equation (5) with two additional triple interactions, $(Treated_m \times Post_t \times RegionalShare_{mt})$ and $(Treated_m \times Post_t \times NationalShare_{mt})$, together with the full set of lower-order interactions implied by the inclusion of $RegionalShare_{mt}$ and $NationalShare_{mt}$. Note that the share of local retailers is omitted as the baseline group.

The results are presented in Table H.10.¹⁴ The main results remain stable across all three inflationary regimes: pooled, inflationary, and deflationary. The triple interactions in the pooled period

¹¹This hypothesis is consistent with [Daruich and Kozlowski \(2023\)](#), finding that prices in stores of multi-region chains respond less to local shocks than those in one-region chains. The hypothesis also aligns with the uniform pricing puzzle documented in [DellaVigna and Gentzkow \(2019\)](#), showing that national chains charge geographically uniform prices and their prices are insensitive to local demand shocks.

¹²[Chenarides et al. \(2024\)](#) find that the entry of a hard discounter reduced markups for incumbent retailers in local markets. Related to it, [Basker and Noel \(2009\)](#) also show that the entry of Walmart reduced the prices of local competitors.

¹³Note that within the egg market during our sample period, we find a positive correlation between HHI and the share of national retailers.

¹⁴Given space constraints, we report only the coefficients on the terms of primary interest.

Table H.10: Triple Difference Estimator (Retailer Composition)

	Inflation	Inflation	Inflation
Bird Flu $\times$ Post $\times$ HHI	0.070* (0.038)	0.096** (0.044)	0.053 (0.082)
Bird Flu $\times$ Post	0.065 (0.105)	-0.043 (0.198)	0.026 (0.119)
HHI $\times$ Post	-0.010* (0.006)	-0.018** (0.009)	0.012 (0.012)
Bird Flu $\times$ HHI	-0.164 (0.138)	-0.197 (0.181)	-0.085 (0.200)
HHI	0.034 (0.026)	0.056* (0.029)	0.022 (0.049)
Bird Flu $\times$ Post $\times$ Regional Share	-0.103 (0.182)	0.072 (0.310)	-0.104 (0.215)
Bird Flu $\times$ Post $\times$ National Share	-0.110 (0.104)	0.028 (0.174)	-0.092 (0.155)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Observations	3,145	1,850	1,295

The table reports regression results from equation (5), including additional triple interactions of Bird Flu $\times$ Post with national retailer share and regional retailer share. The coefficients on these triple interactions are reported in the table. Additional interaction controls involving retailer shares are included in the regressions but omitted from the table for brevity. All other descriptions remain the same as in Table 7. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

(Column 1) and the inflationary period (Column 2) reveal a positive and statistically significant association with inflation as before. In the deflationary period (Column 3), the coefficient remains positive, although it loses statistical significance at the 10 percent level. This suggests that the baseline results on the effect of market concentration on the pass-through of shocks are not solely attributable to differences in retailer composition across MSAs.

Relatedly, a remaining concern is that the retail chains operating in high- and low-concentration MSAs differ systematically along dimensions beyond whether they are a local, regional, or national retailer. Specifically, retailers may differ along unobserved dimensions beyond the characteristics we observe, and such compositional differences could still drive our results rather than market concentration itself.¹⁵ Table H.10 partially addresses this concern by controlling for whether retailers

¹⁵Kim and Navarrete (2026) find that the differences in retailer composition can account for some of the variation in price levels across markets, even for the same product.

Table H.11: Triple Difference Estimator (Eggs, UPC-Retailer Level)

	$\Delta \ln \text{ Price}$	$\Delta \ln \text{ Price}$	$\Delta \ln \text{ Price}$
Bird Flu $\times$ Post $\times$ HHI	0.020*** (0.007)	0.025** (0.011)	0.006 (0.016)
Bird Flu $\times$ Post	-0.015*** (0.004)	-0.004 (0.006)	-0.026** (0.011)
HHI $\times$ Post	-0.011*** (0.003)	-0.005 (0.008)	-0.018** (0.009)
Bird Flu $\times$ HHI	-0.035 (0.035)	-0.053 (0.068)	-0.043 (0.121)
HHI	0.029*** (0.011)	0.050*** (0.018)	0.002 (0.031)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Retailer FE	Yes	Yes	Yes
Observations	258,553	148,728	109,729

Note: The table reports estimates from equation (6) augmented with retailer fixed effects. The unit of observation changes from UPC-MSA-quarter to UPC-retailer-MSA-quarter. The coefficient of interest is the triple interaction of Bird Flu, Post, and HHI, as in equation (6). Bird Flu is a binary variable equal to one for MSAs in which egg farmers culled their layers during the 2014–2015 bird flu episode, and Post is a binary variable equal to one in the post-shock period beginning 2014Q4. HHI is the Herfindahl-Hirschman index of retail chains' egg sales within an MSA, fixed at its 2014Q3 value for all subsequent quarters. The sample period ranges from 2012Q4 to 2016Q4. Inflationary and deflationary periods are determined by the national price index of eggs. Column 1 pools all periods together, Column 2 restricts to the inflationary period, and Column 3 restricts to the deflationary period. MSA, quarter, retailer, and product (UPC) fixed effects are included in all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

are regional, national, or local retailers. To further isolate differences in pricing across low- and high-concentration markets from differences across retailers, we augment our baseline specification in equation (6) by including retailer fixed effects in the product-retailer-MSA-quarter-level analysis.

Table H.11 reports estimates from specifications that include retailer fixed effects. The results are similar in magnitude and statistical significance to our baseline estimates in Table 8, which do not include retailer fixed effects. In the pooled period, we find that retailers in more concentrated markets increased relative prices by about 2 percent more, significant at 1 percent level. The effect is larger during the inflationary period, when relative prices increase by 2.5 percent more in more concentrated markets, significant at the 5 percent level. We find no statistically significant

differential response during the deflationary period. This suggests that the baseline patterns remain robust even after controlling for differences in retailer composition.

H.7 Commodity Cost Shocks in Coffee Market

We provide consistent evidence in the coffee market by exploiting fluctuations in commodity prices, following [Sangani \(2024\)](#). Given that the United States does not produce a meaningful amount of coffee, we treat changes in the commodity price of Arabica coffee as exogenous to U.S. retail conditions. We measure the global price of coffee (Arabica) using the IMF Commodity Database.¹⁶ Our identification strategy exploits time-series variation in global coffee prices. To control for differences in basket composition, we conduct the analysis at the UPC-quarter-MSA level and include UPC and MSA fixed effects. Specifically, we estimate the following regression:

$$\begin{aligned}\Delta \log p_{umt} = & \beta_0 \Delta \log c_t + \beta_1 \text{MidHHI}_{mt} + \beta_2 \text{HighHHI}_{mt} \\ & + \beta_3 (\Delta \log c_t \times \text{MidHHI}_{mt}) + \beta_4 (\Delta \log c_t \times \text{HighHHI}_{mt}) \\ & + \delta_m + \delta_u + \varepsilon_{umt},\end{aligned}\tag{H.3}$$

where $\Delta \log p_{umt}$ denotes the change in the log price of UPC u in MSA m between quarters t and $t + 4$. We classify MSAs into terciles based on the distribution of coffee-market HHI in each quarter. As before, the bottom tercile (*LowHHI*) comprises the one-third of MSAs with the lowest HHI and serves as the reference group. $\Delta \log c_t$ corresponds to the log change in the global Arabica coffee price between quarters t and $t + 4$. We estimate the regression with our pooled sample, for periods of positive cost shocks (quarters in which the global coffee price increases between t and $t + 4$, with positive log price changes), and for periods of negative cost shocks (quarters in which the global coffee price weakly decreases over the same horizon, with zero or negative log price changes).

In Table [H.12](#), we report estimates of pass-through from global Arabica coffee prices to U.S. retail coffee prices. For the baseline group (*LowHHI*), a 1 percent increase in the global coffee price raises retail prices by 0.072 percent (Column 1). The estimated pass-through is 0.068 during positive cost shocks (Column 2) and declines to 0.052 during negative cost shocks (Column 3).

¹⁶The IMF Series ID for coffee is PCOFFOTMUSDM.

Table H.12: Coffee Price Pass-through and Retail Market Concentration across MSAs

	$\Delta \ln \text{Price}$	$\Delta \ln \text{Price}$	$\Delta \ln \text{Price}$
$\Delta \log c_t \times \text{HighHHI}$	0.013*** (0.003)	0.028*** (0.006)	-0.010* (0.006)
$\Delta \log c_t \times \text{MidHHI}$	0.006 (0.004)	0.021*** (0.007)	-0.012* (0.007)
$\Delta \log c_t$	0.072*** (0.002)	0.068*** (0.004)	0.052*** (0.004)
HighHHI	-0.005* (0.003)	-0.012*** (0.003)	-0.002 (0.003)
MidHHI	-0.001 (0.002)	-0.005** (0.002)	-0.002 (0.002)
Sample Periods	All	Positive Cost	Negative Cost
MSA FE	Yes	Yes	Yes
UPC FE	Yes	Yes	Yes
Observations	4,696,457	2,724,256	1,971,691

Notes: The table reports regression results from equation (H.3). The coefficient of interest is the interaction of HighHHI and $\Delta \log c_t$. HighHHI is an indicator for the top tercile MSAs, and MidHHI is an indicator for the mid tercile MSAs in the distribution of sales HHI each quarter. The dependent variable is the log price difference of product u in MSA m between quarters t and $t + 4$. Positive cost shock period contains quarters with positive log price changes, and negative cost shock period includes those with zero or negative log price changes. Column 1 pools all periods together, Column 2 only considers the positive-shock period, and Column 3 only considers the negative-shock period. The sample period ranges from 2006Q1 to 2020Q3. All specifications include UPC and MSA fixed effects. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Even in the least concentrated markets, retailers transmit cost declines less completely than cost increases.

We then examine heterogeneity by local concentration through the interaction terms $\Delta \log c_t \times \text{MidHHI}_{mt}$ and $\Delta \log c_t \times \text{HighHHI}_{mt}$. The coefficient on $\Delta \log c_t \times \text{MidHHI}_{mt}$ is statistically insignificant in the pooled sample. This masks offsetting effects across the cost cycle: the coefficient is positive during periods of rising costs (0.021, significant at the 1 percent level) and negative during periods of falling costs (-0.012, significant at the 10 percent level). By contrast, the interaction with HighHHI_{mt} is positive and significant in the pooled sample (Column 1): a 1 percent increase in the global coffee price raises retail prices by an additional 0.013 percent in MSAs in the top tercile of concentration relative to those in the bottom tercile. The same pattern of offsetting effects appears for the top tercile, with larger magnitudes, at 0.028 during rising costs (Column 2, significant at the 1 percent level) and -0.010 during falling costs (Column 3, significant at the 10 percent level).

Taken together, pass-through is stronger in more concentrated retail markets when costs rise, and weaker when costs fall.¹⁷ Because the two interactions carry opposite signs, they compound over a complete cost cycle rather than offset: a 1 percent cost increase followed by a 1 percent cost decrease leaves retail prices in the top tercile 0.038 percent higher relative to the bottom tercile.¹⁸ This provides complementary evidence that transitory cost shocks can generate persistent differences in local price levels, consistent with the pattern we document in the egg market.

H.8 Cost-related Hypothesis

An alternative hypothesis for spatial disparities in inflation revolves around cost differentials. If marginal costs in lower-income areas rise faster than in higher-income areas, this can contribute to higher inflation in those regions, irrespective of the market structure of retailers. To test this hypothesis, we use wage data for retail workers from the American Community Survey (ACS) and compare wage levels and growth across MSAs with varying income levels from 2006 to 2020.¹⁹ We combine the ACS wage data with BDS data at the MSA level to control for differences in firm composition across regions.

We estimate the following two regressions to examine wage variation in wage levels and wage growth across MSAs with different income levels:

$$w_{mt} = \beta_0 + \beta_1 Income_{mt} + X'_{mt}\gamma + \delta_t + \varepsilon_{mt} \quad (H.4)$$

$$\Delta \ln w_{mt} = \beta_0 + \beta_1 Income_{mt} + X'_{mt}\gamma + \delta_t + \varepsilon_{mt}, \quad (H.5)$$

where w_{mt} ($\Delta \ln w_{mt}$) is the average wage level (or growth) in the retail sector in MSA m and year t , $Income_{mt}$ is the income per capita of MSA m in year t , and X_{mt} is a vector of MSA-level

¹⁷Note here that the cost shock enters with its sign rather than as a treatment indicator. Thus, in the falling-cost sample, a negative interaction coefficient indicates that higher concentration attenuates the response of prices to cost declines. The analogous prediction in the binary bird-flu specification is a positive coefficient on the triple interaction during the deflationary period.

¹⁸For example, consider a 1 percent cost increase followed by a 1 percent cost decrease. During the rising-cost period, the differential pass-through in the top relative to the bottom HHI tercile raises relative prices by 0.028 percent. During the falling-cost period, the interaction coefficient is -0.010 ; because the cost change is -1 percent, this implies an additional relative price increase of 0.010 percent. The cumulative difference is therefore $0.028 + 0.010 = 0.038$ percent.

¹⁹We restrict our sample to prime-age workers (aged 20-55) that earn more than \$5000 and work at least 40 weeks per year in the retail sector.

Table H.13: Average Wage Levels and Growth in Retail Sector across MSAs

	Wage	Wage	Wage	Wage	Wage	Wage
Income	3.785*** (0.508)	2.871*** (0.382)	3.000*** (0.427)	3.770*** (0.532)	2.371*** (0.293)	2.822*** (0.390)
College Share		2.761*** (0.343)			2.441*** (0.289)	2.780*** (0.343)
Large Firm Share			-2.437*** (0.360)		-1.883*** (0.261)	
Large Estab Share				-0.126 (0.393)		-0.368 (0.319)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,868	2,868	2,868	2,868	2,868	2,868
	$\Delta \ln \text{Wage}$	$\Delta \ln \text{Wage}$	$\Delta \ln \text{Wage}$	$\Delta \ln \text{Wage}$	$\Delta \ln \text{Wage}$	$\Delta \ln \text{Wage}$
Income	-0.020 (0.014)	-0.015 (0.013)	-0.023 (0.014)	-0.023 (0.014)	-0.017 (0.082)	-0.018 (0.014)
Δ College Share		0.082*** (0.008)			0.082*** (0.008)	0.082*** (0.008)
Large Firm Share			-0.008 (0.021)		-0.006 (0.020)	
Large Estab Share				-0.025 (0.028)		-0.025 (0.025)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,580	2,566	2,580	2,580	2,566	2,566

Note: The table presents MSA-level wage regression results from equations (H.4) and (H.5). The dependent variable in the top panel is the average wage (in \$1000), while the bottom panel has the log difference in the average wages within retail sector in an MSA in a given year. The sample period spans from 2006 to 2020 with year fixed effects included. The main independent variable is the MSA-level income per capita (in thousands of \$). In Columns 2, 5, 6, the share of college-educated retailer workers is included in the top panel, and its growth is included in the bottom panel as a control. In Columns 3 and 5, the share of large retailers (%), those with 500 employees or more) is controlled, and in Columns 4 and 6, the share of establishments associated with these large retailers is controlled within an MSA in a given year. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

characteristics, including the share (or growth of the share) of college workers in the retail sector, and the share of large retailers (with 500 or more employees) or establishments associated with large retailers, and δ_t is a year fixed effect.

In Table H.13, the top and bottom panels present the results for the average wage levels and growth, respectively. These results reveal that the average wage level is generally lower in lower-income areas, even after controlling for the composition of skills and the share of large firms or establishments in the retail sector. However, the second panel suggests there are no significant patterns in wage growth across MSAs by income level. While the data are aggregated, they provide

suggestive evidence that retailer wages are neither higher nor growing faster in lower-income areas or areas with a higher share of large retailers, making differential retail labor costs an unlikely explanation for the observed inflation patterns.

I Model Validation

I.1 Market Concentration and Markup

The model predicts that markups are higher in markets with greater concentration. We test this prediction by constructing MSA-level average markups for food products using PriceTrak PromoData provided by NielsenIQ. Specifically, using UPC-level wholesale costs and retail prices, we construct a proxy for the markup for each UPC as the ratio of its retail price to wholesale cost. We then aggregate these UPC-level markups to the MSA level using quantity weights.

Next, we estimate the following regression to examine the MSA-level association between aggregate markups and market concentration:

$$Markup_{mt} = \beta HHI_{mt} + \delta_m + \delta_t + \varepsilon_{mt}, \quad (I.1)$$

where $Markup_{mt}$ denotes the MSA-level average markup constructed as described above, HHI_{mt} is the average quarterly market concentration, measured by the sales-based HHI, in MSA m during year t , and δ_m and δ_t denote MSA and year fixed effects, respectively. The regression sample is annual and spans 2006–2012.²⁰

Table I.1 shows a positive association between MSA-level average markups and market concentration, consistent with the model prediction. The results are also robust when both markups and market concentration are constructed using the NielsenIQ Consumer Panel data.

²⁰Note that PromoData only spans 2006–2012. Also, PromoData contains market-level location information for each UPC sold, which we use to map wholesale cost information to each UPC–MSA pair in the NielsenIQ Retail Scanner data. The resulting matched sample includes only a subset of the MSAs in our baseline sample, reflecting some gaps in geographic coverage and product-location overlap across the two datasets.

Table I.1: MSA-level Markups and Market Concentration

	Markup
HHI	0.054** (0.025)
Observations	233
Year FE	Yes
MSA FE	Yes

Note: The table presents the MSA-level relationship between average markup and market concentration in 2006-2012. The regression includes year and MSA fixed effects. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

I.2 Retailer Market Share and Pass-through

The model predicts that firms with sufficiently large market shares exhibit stronger pass-through of cost shocks. We test this prediction using a triple-difference specification that interacts the treatment variables with retailer-level market share.

Specifically, we partition retailers (r) into four groups based on their market share (within MSA m at quarter t) and define indicator variables, $\mathbb{I}_{mrt} = 1, 2, 3, 4$, for each group. We consider two types of classifications. The first is based on market-share quartiles. The second is based on absolute market-share thresholds: $\mathbb{I}_{mrt} = 1$ if market share is below 0.1, $\mathbb{I}_{mrt} = 2$ if market share is between 0.1 and 0.5, $\mathbb{I}_{mrt} = 3$ if market share is between 0.5 and 0.9, and $\mathbb{I}_{mrt} = 4$ if market share exceeds 0.9. Note that we fix it to 2014Q3 value for the post-shock period.

With this, we run the following regression:

$$\begin{aligned}
\Delta \ln price_{umrt} = & \beta_0 + \beta_1(Treated_m \times Post_t) + \sum_{i=1}^4 \left[\beta_{i+1} \mathcal{I}(\mathbb{I}_{mrt} = i) + \beta_{i+5}(Treated_m \times \mathcal{I}(\mathbb{I}_{mrt} = i)) \right. \\
& \left. + \beta_{i+9}(Post_t \times \mathcal{I}(\mathbb{I}_{mrt} = i)) + \beta_{i+13}(Treated_m \times Post_t \times \mathcal{I}(\mathbb{I}_{mrt} = i)) \right] \\
& + \delta_m + \delta_r + \delta_t + \delta_u + \varepsilon_{umrt},
\end{aligned} \tag{I.2}$$

where $\mathcal{I}(\cdot)$ is an indicator function, and δ_r is a retailer fixed effect.

Tables I.2 and I.3 report the results using the market-share quartile and absolute market-share classifications, respectively. Across both specifications, the estimated differential pass-through is positive and statistically significant only for retailers with sufficiently large market shares. This

Table I.2: Triple Difference Estimator by Market Share Quartiles

	$\Delta \ln \text{ Price}$ All	$\Delta \ln \text{ Price}$ Inflation	$\Delta \ln \text{ Price}$ Deflation
Bird Flu $\times$ Post $\times \mathbb{I}_{mrt} = 2$	-0.006* (0.003)	-0.003 (0.004)	-0.004 (0.012)
Bird Flu $\times$ Post $\times \mathbb{I}_{mrt} = 3$	-0.001 (0.007)	-0.002 (0.007)	0.000 (0.012)
Bird Flu $\times$ Post $\times \mathbb{I}_{mrt} = 4$	0.014*** (0.004)	0.015*** (0.005)	0.010 (0.015)
Bird Flu $\times$ Post	-0.009*** (0.003)	0.005 (0.004)	-0.028** (0.011)
$\mathbb{I}_{mrt} = 2 \times \text{Post}$	0.004** (0.002)	-0.010*** (0.002)	0.023*** (0.004)
$\mathbb{I}_{mrt} = 3 \times \text{Post}$	0.006*** (0.002)	-0.020*** (0.003)	0.036*** (0.004)
$\mathbb{I}_{mrt} = 4 \times \text{Post}$	0.003* (0.002)	-0.025*** (0.003)	0.037*** (0.005)
Bird Flu $\times \mathbb{I}_{mrt} = 2$	0.001 (0.005)	-0.002 (0.004)	-0.002 (0.009)
Bird Flu $\times \mathbb{I}_{mrt} = 3$	-0.001 (0.005)	-0.002 (0.006)	-0.001 (0.007)
Bird Flu $\times \mathbb{I}_{mrt} = 4$	-0.013*** (0.003)	-0.014*** (0.003)	-0.009 (0.009)
$\mathbb{I}_{mrt} = 2$	-0.003** (0.001)	0.001 (0.002)	-0.014*** (0.003)
$\mathbb{I}_{mrt} = 3$	-0.004*** (0.001)	0.006*** (0.002)	-0.023*** (0.003)
$\mathbb{I}_{mrt} = 4$	-0.001 (0.001)	0.010*** (0.001)	-0.024*** (0.004)
Sample Period	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Product FE	Yes	Yes	Yes
Retailer FE	Yes	Yes	Yes
Observations	231,582	142,845	88,649

Note: The table reports estimates from a triple difference-in-differences specification interacting the bird flu shock, the post period, and retailer market share quartiles (fixed to 2014Q3 values for all quarters in the post period with $\mathbb{I}_{mrt} = 1$ as the base group). All regressions include quarter, MSA, product, and retailer fixed effects. Standard errors, reported in parentheses, are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

finding is consistent with the model's prediction that pass-through is incomplete and increases among firms with sufficiently large market shares.

Table I.3: Triple Difference Estimator by Market Share Ranges

	$\Delta \ln \text{ Price}$ All	$\Delta \ln \text{ Price}$ Inflation	$\Delta \ln \text{ Price}$ Deflation
Bird Flu $\times$ Post $\times \mathbb{I}_{mrt} = 2$	0.006 (0.005)	-0.001 (0.005)	0.009 (0.009)
Bird Flu $\times$ Post $\times \mathbb{I}_{mrt} = 3$	0.015*** (0.005)	0.020** (0.008)	-0.001 (0.015)
Bird Flu $\times$ Post $\times \mathbb{I}_{mrt} = 4$	0.016*** (0.005)	0.021*** (0.006)	0.003 (0.014)
Bird Flu $\times$ Post	-0.013*** (0.003)	0.004 (0.004)	-0.030*** (0.007)
$\mathbb{I}_{mrt} = 2 \times \text{Post}$	0.003** (0.001)	-0.013*** (0.002)	0.022*** (0.003)
$\mathbb{I}_{mrt} = 3 \times \text{Post}$	0.001 (0.002)	-0.021*** (0.003)	0.026*** (0.004)
$\mathbb{I}_{mrt} = 4 \times \text{Post}$	0.004 (0.002)	-0.025*** (0.005)	0.038*** (0.006)
Bird Flu $\times \mathbb{I}_{mrt} = 2$	-0.004* (0.002)	-0.003 (0.004)	-0.003 (0.006)
Bird Flu $\times \mathbb{I}_{mrt} = 3$	-0.015*** (0.003)	-0.017*** (0.003)	0.005 (0.011)
Bird Flu $\times \mathbb{I}_{mrt} = 4$	-0.010*** (0.003)	-0.008*** (0.003)	-0.007 (0.009)
$\mathbb{I}_{mrt} = 2$	-0.002* (0.001)	0.006*** (0.001)	-0.015*** (0.002)
$\mathbb{I}_{mrt} = 3$	0.001 (0.001)	0.011*** (0.001)	-0.018*** (0.003)
$\mathbb{I}_{mrt} = 4$	-0.003* (0.002)	0.007*** (0.003)	-0.025*** (0.004)
Sample Period	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Product FE	Yes	Yes	Yes
Retailer FE	Yes	Yes	Yes
Observations	231,582	142,845	88,649

Note: The table reports estimates from a triple difference-in-differences specification interacting the bird flu shock, the post period, and retailer group dummies based on market share ranges (fixed to 2014Q3 values for all quarters in the post period with $\mathbb{I}_{mrt} = 1$ as the base group). All regressions include quarter, MSA, product, and retailer fixed effects. Standard errors, reported in parentheses, are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

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